Constrained Energy Minimization (CEM) for Hyperspectral Target Detection: Theory and Generalizations

Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— Target detection is a fundamental task of hyperspectral imaging where constrained energy minimization (CEM) has been widely used for subpixel target detection techniques. Due to its effectiveness, CEM has been generalized to various versions, such as kernel CEM (KCEM), kernel target-constrained interference-minimized filter (KTCIMF), ensemble cascaded CEM (ECEM), and hierarchical CEM (HCEM). Unfortunately, these generalizations overlooked the key design rationale behind CEM. This article revisits CEM for hyperspectral target detection (HTD) and proves how and why it works mathematically. Specifically, several new CEM generalizations are derived and particularly noteworthy. By including spatial information in an iterative process, KCEM, ECEM, and HCEM can be generalized to iterative KCEM (IKCEM), iterative KTCIMF (IKTCIMF), iterative ECEM (IECEM), and iterative HCEM (IHCEM). Also, by utilizing an iterative random training sampling (IRTS) to generate the desired target signature to be detected, these algorithms are further generalized to IRTS KCEM (IRTS-KCEM), IRTS ECEM (IRTS-ECEM), and IRTS HCEM (IRTS-HCEM). A comprehensive analysis along with comparative study on these generalizations is conducted through extensive experiments to demonstrate the effectiveness of IKCEM, IHCEM, and IECEM.**

***Index Terms*— Constrained energy minimization (CEM), ensemble cascaded CEM (ECEM), hierarchical CEM (HCEM), iterative CEM (ICEM), iterative random training sampling CEM (IRTS-CEM), kernel CEM (KCEM), target-constrained interference-minimized filter (TCIMF).**

NOMENCLATURE

AUC	Area under curve.
BA-TCIMF	BKG-annihilated TCIMF.
BKG	Background.
BS	BKG suppression.
CDA	Component decomposition analysis.
CEM	Constrained energy minimization.
DS	Data sphering.
ECEM	Ensemble cascaded CEM.
GT	Ground truth.
HCEM	Hierarchical CEM.
HTD	Hyperspectral target detection.
ICEM	Iterative CEM.
IECEM	Iterative ECEM.
IHCEM	Iterative HCEM.
IKCEM	Iterative KCEM.
IKTCIMF	Iterative KTCIMF.
ILCMV	Iterative LCMV.
INCEM	Iterative NCEM.
IRTS	Iterative random training sampling.
KCEM	Kernel CEM.
KTCIMF	Kernel TCIMF.
KLCMV	Kernel LCMV.
LCMV	Linearly constrained minimum variance.
LSOSP	Least-squares OSP.
MV	Minimum variance.
NCEM	Nonlinear CEM.
ODP	Overall detection probability.
OSP	Orthogonal subspace projection.
ROC	Receiver operating characteristic.
RTS	Random training sampling.
SBR	Signal-to-BKG ratio.
SNR	Signal-to-noise ratio.
SNPR	Signal-to-noise probability ratio.
TCIMF	Target-constrained interference-minimized filter.
TD	Target detectability.

Manuscript received 26 May 2024; revised 25 June 2024; accepted 3 July 2024. Date of publication 5 July 2024; date of current version 18 July 2024. The work of Chein-I Chang was supported in part by the YuShan Fellow Program sponsored by the Ministry of Education in Taiwan and in part by the National Science and Technology Council (NSTC) under Grant 111-2634-F-006-012. *(Corresponding author: Chein-I Chang.)*

The author is with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China, also with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, Baltimore, MD 21250 USA, and also with the Department of Electrical Engineering, National Cheng Kung University, Tainan 70101, Taiwan (e-mail: cchang@umbc.edu).

Digital Object Identifier 10.1109/TGRS.2024.3424281

I. INTRODUCTION

HTD plays a crucial role in hyperspectral data exploitation [1], [2] and can be carried out by target detection as active detection [3] and anomaly detection as passive detection [4]. An earlier developed hyperspectral target detector was the one developed by Harsanyi and Chang [5], called OSP given by

$$\delta^{\mathrm{OSP}}(\mathbf{r}) = \mathbf{d}^T P_{\mathbf{U}}^{\perp} \mathbf{r} \tag{1}$$

where $\mathbf{d}$ is the desired target signature to be detected and $\mathbf{U}$ is a set of the undesired target signatures and an undesired signature projection matrix $P_{\mathbf{U}}^{\perp}$ given by

$$P_{\mathbf{U}}^{\perp} = \mathbf{I} - \mathbf{U}\mathbf{U}^{\#} = \mathbf{I} - \mathbf{U}\left(\mathbf{U}^T\mathbf{U}\right)^{-1}\mathbf{U}^T \tag{2}$$

which annihilates $\mathbf{U}$ prior to target detection. However, in order for OSP to work effectively, the desired target signature $\mathbf{d}$ must

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

be precise and the knowledge of the undesired target signatures **U** also needs to be known a priori. In reality, obtaining **d** may not be difficult, but finding reliable knowledge of **U** is indeed an issue. The CEM [6], [7], [8] was particularly designed to address this issue.

A. Historical Review

The idea of CEM can be traced back to Harsanyi's [6] doctoral dissertation. which made an attempt to address the issue of **U**. Its key idea of CEM was originated from the classic minimum variance distortionless response (MVDR) beamformer, which is a widely used optimal beamforming design to minimize the variance of the beamformer output [9]. If the noise is additive, then the variance of the captured signals is the sum of the variances of the desired signal and the noise. Hence, MVDR seeks a solution to minimize this sum, thereby mitigating the effect of the noise. On the other hand, the well-known multiple signal classification (MUSIC) algorithm is a method for estimating the direction of arrival of signals received by an antenna array [10]. The algorithm inputs the covariance matrix of the signals received by each antenna element, performs an eigendecomposition, and separates the result into signal and noise subspaces. It further derives an angular spectrum and finally searches that spectrum for peaks that represent signal arrival angles. CEM was developed to combine the MUSIC and MVDR filter to derive a hyperspectral target detector that constrains the desired target signature interpreted as the desired direction of arrival (DOA) while minimizing the minimum variance (MV) resulting from undesired signatures **U** interpreted as undesired DOA via the MVDR filter. As a consequence, there is no need of knowing undesired signal signatures, **U**. In this case, CEM only needs to know the desired target signature **d** while discarding **U**. Other than that no knowledge is required. Such a significant advantage makes CEM practically implementable and a valuable technique in real-world applications because it does not require BKG knowledge.

B. Motivation

No techniques can be claimed to be perfect or universally optimal. CEM is not exceptional. A technique generally has its strengths and also weakness. How to take advantage of its strengths and in the meantime minimizing its weakness is the rule of thumb to best utilize the technique. For CEM to be effective, its strengths are its BKG suppressibility by inverting the global sample correlation matrix **R**. It only requires the knowledge of a target to be detected **d** without full target knowledge. However, these strengths also traded for its weakness that the CEM performance is also very sensitive to its used **R** and **d**. Another weakness is the inherent linearity of CEM, which limits CEM to dealing with nonlinearity separability effectively. Therefore, in order to improve CEM, we need to seek solutions to overcome its weaknesses.

One is that CEM is a pixel-based spectral detector and does not include spatial information. To address this issue, Xue et al. [11] were the first work to develop an ICEM that implements a set of spatial filters to capture spatial information from CEM-detection maps via an iterative process so as to improve hyperspectral image classification (HSIC). The resulting ICEM is called ICEM that is actually a spectral–spatial target detector. Recently, Yang et al. [12] developed a spectral–spatial CEM, which is a constrained detection algorithm, referred to as sparse-spatial CEM, to simultaneously force the sparsity and spatial correlation of the detection output via proper regularizations.

Another issue is how to deal with the required knowledge of **d**. In general, the true knowledge of **d** is obtained by either GT provided by prior knowledge or visual inspection. One way to alleviate this issue is to use training samples to calculate the desired **d** developed by Chang et al. [13]. This leads to the development of training sampling CEM for HSIC [14]. Its idea can be used to extend CEM to IRTS CEM (IRTS-CEM), which randomly selects a set of training samples to calculate **d**. It then uses the obtained **d** to perform CEM and feeds the spatial information obtained from the CEM-detection map back to expand the current dataset to a new data cube from which a new set of training samples will be randomly selected to calculate **d** again to perform CEM. The same process is repeated over again in an iterative manner. By virtue of such iteratively expanded data cubes, the training samples can be updated by including new generated spatial information from CEM-detection maps. Also, by means of random sampling, the training samples can be randomly reselected as many data samples as possible iteratively to cover the GT. This RTS strategy has been very effective to improve spectral spatial classification [15].

Despite that CEM has been shown very effective in subpixel target detection, its performance is constrained on its linear finite impulse response (FIR) filter. Accordingly, it is not fair to compare CEM against a nonlinear target detector. Therefore, the first solution is to relax the linearity constraint imposed on CEM. To make CEM capable of dealing with nonlinear separability issues, a version of CEM using a nonlinear kernel, called KCEM, was first developed by Jiao and Chang [16]. Unfortunately, due to the excessive computational complexity of kenerlization on all data samples used by the global sample correlation matrix, **R**, such KCEM has not received its deserved attention in the past. In order to mitigate this computational dilemma, several approaches have been recently investigated. One is called HCEM developed by Zou and Shi [17], which expanded CEM in a hierarchical layer and kernelized the CEM output by a nonlinear kernel at each layer. It turns out that with such a simple kernelization on the CEM output detection maps, HCEM has shown great improvement, specifically, on BS caused by nonlinearity target separability. However, HCEM did not take full advantage of kernelizing all data samples. In another approach, CEM was extended by Zhao et al. [18] to ECEM, which improved the detection of CEM over nonlinearity and generalization ability. It consists of two-stage processes, multiscale scanning process, and a cascade of CEM detectors. Unfortunately, ECEM only kernelizes feature vectors, not all data samples. A third approach revisited KCEM and made use of the Nyström method to ease the computational burden on kernelizing all the data samples to derive NCEM [19]. With the use of all kernelized data samples, NKCEM can be shown to perform as well as HCEM and ECEM.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

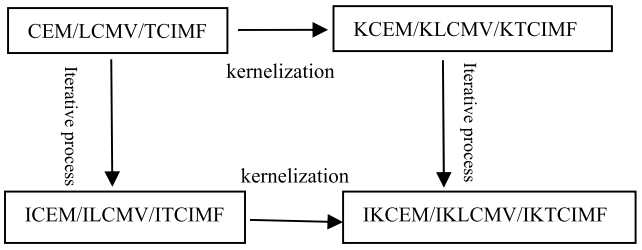


Fig. 1. Diagram of a four-way relationship among CEM/LCMV/TCIMF, KCEM'KLCMV/KTCIMF, ICEM/ILCMV/ITCIMF, and IKCEM/IKLCMV/IKTCIMF.

In the original work [6], CEM was developed to detect a single signal source. When it comes to detect multiple targets, CEM must be implemented multiple times to detect one target at a time. To resolve this issue, Frost [20] extended CEM to LCMV filter, which utilizes multiple linear constraints to replace a single constraint where each constraint is specified to respond one particular signal source. The resulting detector is called LCMV target detector. Moreover, when there is additional knowledge about unwanted signal sources, $\mathbf{U}$ for annihilation, combining LCMV with OSP via (2) leads to a TCIMF developed by Ren and Chang [21]. Also, in analogy with ICEM, an iterative process can also be incorporated into LCMV and TCIMF to derive ILCMV in [22] and iterative TCIMF (ITCIMF) [23]. Similar to KCEM, including kernels in LCMV and TCIMF yields KLCMV [19] and KTCIMF [24]. Finally, KTCIMF was further extended to BKG-annihilated KTCIMF (BA-KTCIMF), which implements two stages of BS, global BS and target BS in KTCIMF [25]. Fig. 1 summarizes a four-way relationship among CEM, KCEM, ICEM, IKC.ILCMV/ITCIMF, and IKCEM/IKLCMV/IKTCIMF.

C. Contributions

The major contributions of this article can be summarized as follows.

1) Although CEM has been widely used for hyperspectral subpixel target detection, its design idea and mathematical theory have been overlooked. This article offers insight into the design rationale of CEM and provides a proof of how CEM is derived mathematically. It also proved that the MV, i.e., minimum output energy in least-squares (LS) sense resulting from CEM, is exactly the reciprocal of SBR. This makes sense because the higher the SBR is, the smaller the LS error in the output energy.
2) In addition, this article also shows why CEM works mathematically, a fact that many people have taken it for granted but never explored its potential and promise.
3) Specifically, this article derives a new version of CEM, called SBR-based CEM (SBR-CEM). Most importantly, it explores one-to-one correspondence between SBR-CEM using $\mathbf{R}^{-1}$ and OSP given by (1) using $P_{\mathbf{U}}^{\perp}$, a fact that has never been realized in the past.
4) Furthermore, this article also generalizes CEM to several extended versions, KCEM, HCEM and ECEM to IKCEM, iterative HECEM (IHCEM) and IECEM and IRTS versions, IRTS-KCEM, IRTS-HCEM and IRTS-ECEM, all of which are new and have not been investigated.

The remainder of this article is organized as follows. Section II provides the design idea and a mathematical proof of CEM. Section III develops a new version of CEM, SBR-based CEM. Section IV shows how CEM works. Section V generalizes CEM to various versions. Section VI derives eight detection measures obtained from 3-D ROC curve analysis. Section VII describes two real hyperspectral images to be used for experiments. Section VIII discusses the experimental results. Section IX draws conclusions.

II. Mathematical Theory of CEM

CEM was originally developed to detect targets at the subpixel level. Its idea arises from the linearly adaptive beamforming [9] by identifying signal arriving from one specific direction as the desired signal source $\mathbf{d}$ via designing a linear FIR filter to preserve the desired signal while minimizing the output energy of signal sources from other undesired directions.

Suppose that a hyperspectral image is represented by a collection of image pixel vectors, denoted by $\{\mathbf{r}_i\}_{i=1}^{N}$ with $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ being an L-dimensional pixel vector and L being the total number of spectral bands. More specifically, let original hyperspectral image (HSI) sample space be Ω_{data}, where $\Omega_{\text{data}} = \{\mathbf{r}_n\}_{n=1}^{N}$ and $\mathbf{r}_n \in \Re^{L\times 1}$ are the nth data sample with L bands. Let $\delta(\mathbf{r}_n)$ be a linear filter specified by a weighting vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$ as

$$\delta(\mathbf{r}_n) = \mathbf{w}^T\mathbf{r}_n. \tag{3}$$

Then, its averaged output energy over the entire image scene is represented by

$$\frac{1}{N}\sum_{n=1}^{N}\delta^2(\mathbf{r}_n) = \mathbf{w}^T\mathbf{R}\mathbf{w} \tag{4}$$

where $\mathbf{R} = (1/N)\sum_{n=1}^{N}\mathbf{r}_n\mathbf{r}_n^T \in \Re^{L\times L}$ is the sample autocorrelation matrix, N is the total number of data samples, and L is the number of spectral bands. Now, we can formulate the following constrained optimization problem as follows:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\} \quad \text{s.t. } \mathbf{d}^T\mathbf{w} = 1 \tag{5}$$

which is the MV of (3) and is actually LS output energy of (4).

Using a Lagrange multiplier λ, we minimize the following constrained problem:

$$J(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{R}\mathbf{w} + \lambda(\mathbf{d}^T\mathbf{w} - 1). \tag{6}$$

Differentiating (4) with respect to $\mathbf{w}$ and λ, we have

$$\left.\frac{\partial J(\mathbf{w}, \lambda)}{\partial \mathbf{w}}\right|_{\mathbf{w}^{\text{CEM}}} = 2\mathbf{R}\mathbf{w}^{\text{CEM}} + \lambda\mathbf{d} = 0 \Rightarrow \mathbf{w}^{\text{CEM}} = -\frac{\lambda}{2}\mathbf{R}^{-1}\mathbf{d} \tag{7}$$

and

$$\left.\frac{\partial J(\mathbf{w}, \lambda)}{\partial \lambda}\right|_{\lambda^{\text{CEM}}} = \mathbf{d}^T\mathbf{w}^{\text{CEM}} = 1. \tag{8}$$

Using (8) we reexpress (7) as

$$\mathbf{d}^T\mathbf{w}^{\text{CEM}} = -\frac{\lambda^{\text{CEM}}}{2}\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} = 1 \Rightarrow \lambda^{\text{CEM}} = -\frac{2}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

By substituting (9) into (7), we can derive

$$\mathbf{w}^{\mathrm{CEM}} = \mathbf{R}^{-1}\mathbf{d}(\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d})^{-1} \tag{10}$$

Substituting (10) into (3) yields the CEM detector, $\delta^{\mathrm{CEM}}(\mathbf{r})$ developed in [1.6, 8] which is given by

$$\delta^{\mathrm{CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{r} = \frac{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{11}$$

Also, substituting (10) into (5) yields MV solving (5) given by

$$\mathrm{MV}\left(\delta^{\mathrm{CEM}}\right) = \left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{R}\mathbf{w}^{\mathrm{CEM}} = \frac{1}{\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}}. \tag{12}$$

Theorem 1:

$$\mathrm{SNR}\left(\delta^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} = \left(\mathrm{MV}\left(\delta^{\mathrm{CEM}}\right)\right)^{-1}. \tag{13}$$

Proof: Let $\overline{\mathbf{X}}$ be the whitened data sample space obtained by applying a whitened matrix $\mathbf{R}^{-1/2}$ to the original data space $\mathbf{X}$ via

$$\overline{\mathbf{r}}_n = \mathbf{R}^{-1/2}\mathbf{r}_n \tag{14}$$

which removes the first two orders of data statistics, mean, and variance/covariance. The linear filter (3) is then operated on $\overline{\mathbf{X}}$ via (14) to yield

$$\delta_{\mathbf{w}}(\overline{\mathbf{r}}_n) = \mathbf{w}^T\overline{\mathbf{r}}_n \tag{15}$$

and (10) becomes $\overline{\mathbf{w}}^{\mathrm{CEM}} = \overline{\mathbf{d}}^T$. Then, its SNR in $\overline{\mathbf{X}}$ is calculated by

$$\begin{aligned}\mathrm{SNR}\left(\delta^{\mathrm{CEM}}(\overline{\mathbf{r}}_n)\right) &= \frac{\left(\overline{\mathbf{w}}^{\mathrm{CEM}}\right)^T\overline{\mathbf{r}}_n\left(\left(\overline{\mathbf{w}}^{\mathrm{CEM}}\right)^T\overline{\mathbf{r}}_n\right)^T}{\left[E\left(\left(\mathbf{w}^{\mathrm{CEM}}\right)^T\overline{\mathbf{n}}\right)^2\right]} \\ &= \frac{\left(\overline{\mathbf{d}}^T\overline{\mathbf{r}}_n\right)^2}{\left[E\left(\overline{\mathbf{d}}^T\overline{\mathbf{n}}\right)^2\right]} \\ &\le \frac{\left(\overline{\mathbf{d}}^T\overline{\mathbf{d}}\right)^2}{\left[E\left(\overline{\mathbf{d}}^T\overline{\mathbf{n}}\right)^2\right]} \text{ (by Schwarz's inequality)}\end{aligned} \tag{16}$$

where $\overline{\mathbf{n}}$ is the noise in $\overline{\mathbf{X}}$ and the equality in (16) holds if and only if

$$\overline{\mathbf{r}}_n^* = \kappa\overline{\mathbf{d}} \tag{17}$$

for any constant κ by Schwarz's inequality.

Substituting $\mathbf{w}^*$ in (17) into (16) results in the maximal SNR given by

$$\mathrm{SNR}(\delta_{\mathbf{w}^*}) = \kappa\overline{\mathbf{d}}^T\overline{\mathbf{d}} = \kappa\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} \tag{18}$$

where the scalar constant κ vanishes since it is canceled out in (16). When $\kappa = 1$ in (18), it is reduced to the SNR of CEM

$$\mathrm{SNR}\left(\delta^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} \tag{19}$$

which is exactly (13). ■

The implication of Theorem 1 indicates that the higher the SNR, the smaller the MV. This makes sense.

Now, if we let Δ be all data samples specified by the target signature $\mathbf{d}$, then we can define a new correlation matrix $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}$ by removing all data samples in Δ from the entire data sample space as

$$\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta} = \frac{1}{|\Omega_{\mathrm{data}} - \Delta|}\sum_{\Omega_{\mathrm{data}}-\Delta}\mathbf{r}_n\mathbf{r}_n^T \tag{20}$$

where $|\Omega_{\mathrm{data}} - \Delta|$ is the size of $\Omega_{\mathrm{data}} - \Delta$.

In the case of (5), $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}$ is formed by all data samples not specified by $\mathbf{d}$ but considered as BKG samples

$$\mathrm{SBR}\left(\delta_{\mathbf{d}}^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{d} \tag{21}$$

which can be interpreted as signal ($\mathbf{d}$)-to-BKG (Δ) ratio (SBR) of $\delta_{\mathbf{d}}^{\mathrm{CEM}}$. Replacing $\mathbf{R}^{-1}$ in (5) with $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}$ in (20) gives rise to

$$\left(\mathbf{w}^{\mathrm{CEM}}\right)^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{w}^{\mathrm{CEM}} = \left(\mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{d}\right)^{-1} \tag{22}$$

which implies that MV of (5) is indeed the reciprocal of the SBR in (21). This is also confirmed by [3, eq. (50)] and [26], which is SNR derived by the R-SNR approach and also defined in [27, p. 57].

Since inverting the correlation matrix $\mathbf{R}$ is simply equivalent to suppressing the interfering effects of data samples that form $\mathbf{R}$ on detection of $\mathbf{d}$. Therefore, technically speaking, CEM should use (12) instead of $\mathbf{R}$ to have better target detection performance as shown in [1]. However, due to the fact that the size of Δ is generally very small, the difference between $\mathbf{R}$ and (20) is not much. This is why CEM has been using $\mathbf{R}^{-1}$ rather than $\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}$. In this case, (21) is

$$\mathrm{SBR}\left(\delta_{\mathbf{d}}^{\mathrm{CEM}}\right) = \mathbf{d}^T\mathbf{R}_{\Omega_{\mathrm{data}}-\Delta}^{-1}\mathbf{d} \approx \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d} \tag{23}$$

which further shows that the reciprocal of $\mathrm{MV}(\delta^{\mathrm{CEM}})$ in (12) is actually the $\mathrm{SBR}(\delta^{\mathrm{CEM}})$ in (23).

III. SBR-Constrained Energy Minimization

The commonly used CEM specified by (11) is based on MVDR that minimizes the MV of the filter-out energy resulting from unwanted signal sources while using a constant to constrain the desired signal source. In hyperspectral image and signal processing, BKG is a main issue in causing degradation of performance. Accordingly, SBR is a more appropriate criterion to measure detection performance as proved by Theorem 1. In this case, the constraint in (5) used to derive CEM is replaced by SBR in (13) to derive a new version of CEM, called SBR-based CEM in Section III-A.

A. SBR Based CEM

This section derives an SBR-CEM. Interestingly, from (23), we can replace the constraint $\mathbf{d}^T\mathbf{w} = 1$ with the constraint specified by the SBR $\mathbf{d}^T\mathbf{w} = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}$. Then, (5) becomes a constrained optimization problem n given by

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\} \quad \text{s.t. } \mathbf{d}^T\mathbf{w} = \mathrm{SBR} = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}. \tag{24}$$

The solution to (24) is called SBR-CEM, which can be provided in the following theorem.

Theorem 2:

$$\delta_{\mathbf{d}}^{\text{SBR-CEM}}(\mathbf{r}) = \left(\mathbf{w}^{\text{SBR-CEM}}\right)^T\mathbf{r} = \mathbf{d}^T\mathbf{R}^{-1}\mathbf{r} \tag{25}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

where subscript "**d**" is included to emphasize the matched signal.

Proof: Introducing a Lagrange multiplier λ into (24) yields

$$J(\mathbf{w},\lambda)=\mathbf{w}^T\mathbf{R}\mathbf{w}+\lambda\left(\mathbf{d}^T\mathbf{w}-\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\right). \tag{26}$$

Differentiating (26) with respect to $\mathbf{w}$ and λ, we have

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \mathbf{w}}\right|_{\mathbf{w}^{\text{SBR-CEM}}}=0\Rightarrow \mathbf{w}^{\text{SBR-CEM}}=-\frac{\lambda}{2}\mathbf{R}^{-1}\mathbf{d} \tag{27}$$

$$\left.\frac{\partial J(\mathbf{w},\lambda)}{\partial \lambda}\right|_{\lambda^{\text{SBR-CEM}}}=0 \Rightarrow \mathbf{d}^T\mathbf{w}^{\text{SBR-CEM}}=\text{SBR}=\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}. \tag{28}$$

Using (27) and (28), we obtain

$$\mathbf{d}^T\mathbf{w}^{\text{SBR-CEM}}=-\frac{\lambda^{\text{CEM}}}{2}\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}=\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}\Rightarrow\lambda^{\text{SBR-CEM}}=-2. \tag{29}$$

Substituting (29) into (27) yields

$$\mathbf{w}^{\text{SBR-CEM}}=\mathbf{R}^{-1}\mathbf{d}. \tag{30}$$

By substituting (30) into (3), the original CEM in (11) becomes SBR-constrained CEM (SBR-CEM), $\delta_{\mathbf{d}}^{\text{SBR-CEM}}(\mathbf{r})$, given by (17). ■

B. Relationship Between CEM and OSP

It is noteworthy to explore the relationships and differences between CEM $\delta^{\text{CEM}}(\mathbf{r})$ in (11) and SBR-CEM, $\delta_{\mathbf{d}}^{\text{SBR-CEM}}(\mathbf{r})$ in (17). The key normalization constant $\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}$ appearing in (11) but absent in (17). If the data sample $\mathbf{r}$ happens to be the desired target signature $\mathbf{d}$, then both achieve different constants, i.e., $\delta^{\text{CEM}}(\mathbf{d})=1$ and $\delta_{\mathbf{d}}^{\text{SBR-CEM}}(\mathbf{d})=\mathbf{d}^T\mathbf{R}^{-1}\mathbf{d}=$ SBR. This indicates that for each data sample vector $\mathbf{r}$, if CEM is used, it can be used as an abundance estimator of $\mathbf{r}$, specifically, as an endmember detector when $\mathbf{d}$ is an endmember. On the other hand, if SBR-CEM is used, it can be used as a target detector to estimate the SBR of $\mathbf{r}$.

Interestingly, the relationship between (11) and (25) noted above can also be found in OSP (1). According to [1] and [5], OSP was derived from SNR. In this case, OSP in (1) is more accurately referred to as SNR-OSP and can be considered as a counterpart of SBR-CEM. Also, from [28] and [29], an LSOSP can be derived as

$$\delta^{\text{LSOSP}}(\mathbf{r})=\frac{\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{r}}{\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{d}} \tag{31}$$

which is an alternative version of OSP in (1) normalized by the constant, $\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{d}$. Now, let the data sample $\mathbf{r}$ be the desired target signature $\mathbf{d}$, and then, (1) and (31) achieve different constants, i.e., $\delta^{\text{LSOSP}}(\mathbf{d})=1$ and $\delta^{\text{OSP}}(\mathbf{d})=\mathbf{d}^T P_{\mathbf{U}}^{\perp}\mathbf{d}=$ SNR, which happens to be SNR in [1], [5], and [27]. This indicates that if LSOSP is used for each data sample vector $\mathbf{r}$, it can be used as an abundance estimator of $\mathbf{r}$, specifically, as an endmember detector when $\mathbf{d}$ is an endmember. On the other hand, if OSP is used, it can be used as a target detector to estimate the SNR of $\mathbf{r}$. Amazingly, the two pairs (CEM, SBR-CEM) and (LSOSP, OSP) are essentially the same with the only difference in how to deal with unknown signatures where CEM suppresses BKG by inverting $\mathbf{R}$, while OSP is used $P_{\mathbf{U}}^{\perp}$ to annihilate the undesired signatures. With the introduction of SBR-CEM, we can complete a four-way relationship among SBR-CEM versus SNR-OSP and CEM versus LSOSP, as shown in Fig. 2. In fact, the MV of (5) is actually a result of LS output energy in (4).

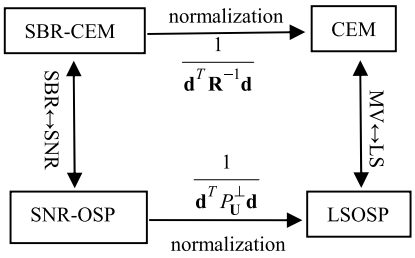


Fig. 2. Diagram of a four-way relationship among CEM, SBR-CEM, SNR-OSP, and LSOSP.

IV. Why CEM Works

In Section II, how CEM is derived was proved mathematically. In this section, we will show why CEM works mathematically.

Assume that $\mathbf{r}=(r_1,r_2,\ldots,r_L)^T$ is an L-dimensional hyperspectral data sample vector and $\{\overline{\lambda}_l\}_{l=1}^L$ are covariance eigenvalues of $\mathbf{R}$ with $\{\overline{\mathbf{v}}_l\}_{l=1}^L$ corresponding to their associated eigenvalues, denoted by orthonormal eigenvectors of length $||\overline{\mathbf{v}}_l||=1$. Then, we can express $\mathbf{R}$ by a spectral decomposition represented by sum of eigenvectors, $\{\overline{\mathbf{v}}_l\}_{l=1}^L$ weighted by their corresponding eigenvalues $\{\overline{\lambda}_l\}_{l=1}^L$ in the following theorem, which is similar to Theorem 7.1 in Chapter 7 with the global autocovariance matrix $\mathbf{K}$ that is replaced by the global autocorrelation matrix $\mathbf{R}$.

Theorem 3:

$$\mathbf{R}=\sum_{i=1}^{L}\overline{\lambda}_i\overline{\mathbf{v}}_i\overline{\mathbf{v}}_i^T. \tag{32}$$

Proof: Since $\{\overline{\lambda}_l\}_{l=1}^L$ and $\{\overline{\mathbf{v}}_l\}_{l=1}^L$ are eigenvalues and their corresponding eigenvectors of $\mathbf{R}\mathbf{r}$

$$\mathbf{R}\overline{\mathbf{v}}_i=\overline{\lambda}_i\overline{\mathbf{v}}_i. \tag{33}$$

Because $\{\overline{\mathbf{v}}_l\}_{l=1}^L$ can be considered as a set of base vectors, any data sample vector $\mathbf{r}$ can be expressed as

$$\mathbf{r}=\sum_{l=1}^{L}r_l\overline{\mathbf{v}}_l\Rightarrow r_l=\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle=\mathbf{r}^T\overline{\mathbf{v}}_l. \tag{34}$$

Substituting (34) into (33) yields

$$\begin{aligned}\mathbf{R}\mathbf{r}&=\mathbf{R}\left(\sum_{l=1}^{L}r_l\overline{\mathbf{v}}_l\right)=\mathbf{R}\left(\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\mathbf{v}}_l\right)=\sum_{i=1}^{L}\mathbf{R}(\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\mathbf{v}}_l)\\&=\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\mathbf{R}(\overline{\mathbf{v}}_l)=\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\lambda}_l\overline{\mathbf{v}}_l=\sum_{l=1}^{L}\overline{\lambda}_l\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\mathbf{v}}_l\\&=\sum_{l=1}^{L}\overline{\lambda}_l\overline{\mathbf{v}}_l\langle\overline{\mathbf{v}}_l,\mathbf{r}\rangle=\sum_{l=1}^{L}\overline{\lambda}_l\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\mathbf{r}=\left(\sum_{l=1}^{L}\overline{\lambda}_l\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\right)\mathbf{r}\end{aligned} \tag{35}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

which implies that

$$\left(\mathbf{R}-\sum_{l=1}^{L}\overline{\lambda}_l\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\right)\mathbf{r}=\mathbf{0}. \tag{36}$$

However, (36) is true for all $\mathbf{r}\neq\mathbf{0}$. Therefore, (32) in Theorem 3 immediately follows Corollary 1. ∎

Corollary 1:

$$\mathbf{R}^{-1}=\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\mathbf{v}_l^T. \tag{37}$$

Proof: Since

$$\mathbf{R}^{-1}\overline{\mathbf{v}}_l=\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l \tag{38}$$

we can use the same treatment carried out in the proof of Theorem 1 to derive

$$\begin{aligned}\mathbf{R}^{-1}\mathbf{r}&=\mathbf{R}^{-1}\left(\sum_{l=1}^{L}r_i\overline{\mathbf{v}}_l\right)=\mathbf{R}^{-1}\left(\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\mathbf{v}}_l\right)\\&=\sum_{i=1}^{L}\mathbf{R}^{-1}(\langle\mathbf{r},\mathbf{v}_i\rangle\mathbf{v}_i)=\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\mathbf{R}^{-1}(\overline{\mathbf{v}}_l)\\&=\sum_{l=1}^{L}\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle\overline{\lambda}_i^{-1}\overline{\mathbf{v}}_l=\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\langle\overline{\mathbf{v}}_l,\mathbf{r}\rangle\\&=\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\mathbf{r}=\left(\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\right)\mathbf{r}\\&\Rightarrow\left(\mathbf{R}^{-1}-\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T\right)\mathbf{r}=\mathbf{0}\\&\Rightarrow\mathbf{R}^{-1}=\sum_{i=1}^{L}\overline{\lambda}_l^{-1}\overline{\mathbf{v}}_l\overline{\mathbf{v}}_l^T.\end{aligned} \tag{39}$$

∎

Using Corollary 1, we can easily derive the following corollary.

Corollary 2:

$$\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}=\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\mathbf{d}^T\overline{\mathbf{v}}_l\mathbf{v}_l^T\mathbf{r}=\sum_{l=1}^{L}\overline{\lambda}_l^{-1}\left(\mathbf{d}^T\overline{\mathbf{v}}_l\right)\left(\mathbf{r}^T\overline{\mathbf{v}}_l\right) \tag{40}$$

where $\mathbf{d}^T\overline{\mathbf{v}}_l=\langle\mathbf{d},\overline{\mathbf{v}}_l\rangle$ and $\mathbf{r}^T\overline{\mathbf{v}}_l=\langle\mathbf{r},\overline{\mathbf{v}}_l\rangle$ are inner products of $\mathbf{d}$ with $\overline{\mathbf{v}}_l$ and $\mathbf{r}$ with $\overline{\mathbf{v}}_l$, respectively, which are projections of $\mathbf{d}$ and $\mathbf{r}$ on the lth eigenvector $\overline{\mathbf{v}}_l$, respectively.

Generally speaking, BKG is made up of a large body of image data, and targets represent relatively a small number of pixels. In this case, the eigenvalues that characterize BKG are generally much greater than the eigenvalues that characterize targets. As a consequence, the reciprocals of larger eigenvalues will be very small, while the reciprocals of smaller eigenvalues become very large. Theorem 2 simply says that the reciprocals of small eigenvalues will dominate the reciprocals of larger eigenvalues. Corollary 2 shows that CEM in (11) and SBR-CEM in (25) are actually determined and dominated by small eigenvalues in (40), which represent targets.

Also, according to Corollary 2, we can interpret CEM as the linear sum of the L eigenvectors $\{\overline{\mathbf{v}}_l\}_{l=1}^{L}$ matched with two specific signals $\mathbf{d}$ and $\mathbf{r}$ weighted by $\{\overline{\lambda}_l^{-1}\}_{l=1}^{L}$. Since $\{\overline{\mathbf{v}}_l\}_{l=1}^{L}$ is an orthonormal basis vector, we further assume that $\overline{\mathbf{v}}_l$ is an lth unity vector, $\overline{\mathbf{v}}_l=(0,\ldots,0,\underbrace{1}_{l},0,\ldots,0)$. Then, a special case of Corollary 2 is

$$\begin{aligned}\mathbf{d}^T\mathbf{R}^{-1}\mathbf{r}&=\mathbf{d}^T\mathbf{\Lambda}^{-1}\mathbf{r}\\&=(d_1,\ldots,d_L)^T\begin{bmatrix}\overline{\lambda}_1^{-1}&0&\cdots&0\\0&\overline{\lambda}_2^{-1}&\ddots&\vdots\\\vdots&\ddots&\ddots&0\\0&\cdots&0&\overline{\lambda}_L^{-1}\end{bmatrix}\begin{pmatrix}r_1\\\vdots\\r_L\end{pmatrix}\\&=\sum_{l=1}^{L}\overline{\lambda}_i^{-1}d_lr_l\end{aligned} \tag{41}$$

where $\mathbf{\Lambda}^{-1}$ is a diagonal matrix of $\mathbf{R}^{-1}$.

V. Various Generalizations of CEM

In anomaly detection, the Reed–Xiaoli anomaly detector (RX-AD) developed by Reed and Yu [30] has been considered as a benchmark anomaly detector. Similarly, in HTD, CEM can also be considered as a benchmark target detector. By virtue of CEM, various generalized CEM versions can also be derived in this section.

A. Nonlinear CEM

1) Kernel CEM: As a linear filter, CEM has its inability in dealing with nonlinear separability problem. To resolve this issue, a general approach is to use a kernel to map data samples from the original space to a higher dimensional feature space. This section generalizes CEM to a kernel version of CEM, called KCEM.

Assume that $\mathbf{X}_{L\times N}$ is a data matrix formed by all data samples in Ω_{data} as $\mathbf{X}_{L\times N}=[\mathbf{r}_1\ \mathbf{r}_2\ \cdots\ \mathbf{r}_N]$ where each column vector represents a data sample. Let φ be a nonlinear function that maps an original data sample vector $\mathbf{r}_n$ into a high-dimensional Hilbert feature space, H via $\varphi:\Re^L\to H$ by $\mathbf{r}_n\to\varphi(\mathbf{r}_n)$ and $\mathbf{X}\to\mathbf{X}_\varphi$ with $\mathbf{X}_\varphi=[\varphi(\mathbf{r}_1)\varphi(\mathbf{r}_2)\ \cdots\ \varphi(\mathbf{r}_N)]$, where the symbol with φ indicates that they are in the feature space H. The new autocorrelation matrix $\mathbf{R}_\varphi$ in the Hilbert space can be computed and expressed as

$$\mathbf{R}_\varphi=\frac{1}{N}\sum_{n=1}^{N}\varphi(\mathbf{r}_n)\varphi(\mathbf{r}_n)^T=\frac{1}{N}\sum_{n=1}^{N}\mathbf{X}_\varphi\mathbf{X}_\varphi^T. \tag{42}$$

Thus, operating $\delta^{\text{CEM}}(\mathbf{r})$ in the Hilbert space results in KCEM given by

$$\begin{aligned}\delta^{\text{KCEM}}(\mathbf{r}_n)&=\left(\mathbf{w}^{\text{KCEM}}\right)^T\varphi(\mathbf{r}_n)\\&=\varphi(\mathbf{d})^T\mathbf{R}_\varphi^{-1}\varphi(\mathbf{r}_n)\left(\varphi(\mathbf{d})^T\mathbf{R}_\varphi^{-1}\varphi(\mathbf{d})\right)^{-1}\end{aligned} \tag{43}$$

where $\mathbf{w}^{\text{KCEM}}$ is given by

$$\mathbf{w}^{\text{KCEM}}=\mathbf{R}_\varphi^{-1}\varphi(\mathbf{d})\left(\varphi(\mathbf{d})^T\mathbf{R}_\varphi^{-1}\varphi(\mathbf{d})\right)^{-1}. \tag{44}$$

Since (43) is impossible to solve due to the infinite dimensionality in the feature space H, an RBF kernel function defined by

$$K(\mathbf{r}_n,\mathbf{r}_j)=\exp\left(-\|\mathbf{r}_n-\mathbf{r}_j\|^2/2\sigma^2\right) \tag{45}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

is used to calculate (43). Using the derivations in [19], the KCEM can be derived as

$$\delta^{\mathrm{KCEM}}(\mathbf{r}_n) = \mathbf{k}_{\mathbf{d}}^T \mathbf{K}^{-2} \mathbf{k}_{\mathbf{r}_n} \left(\mathbf{k}_{\mathbf{d}}^T \mathbf{K}^{-2} \mathbf{k}_{\mathbf{d}} \right)^{-1} \tag{46}$$

where $\mathbf{k}_{\mathbf{r}_n} = (K(\mathbf{r}_n, \mathbf{r}_1), K(\mathbf{r}_n, \mathbf{r}_2), \ldots, K(\mathbf{r}_n, \mathbf{r}_N))^T$ and $\mathbf{k}_{\mathbf{d}} = (K(\mathbf{d}, \mathbf{r}_1), K(\mathbf{d}, \mathbf{r}_2), \ldots, K(\mathbf{d}, \mathbf{r}_N))^T$.

However, computing the kernel matrix $\mathbf{K}$ in (46) is very expensive in terms of computational times and memories. In this case, low-rank approximation such as Nyström method [19] is used to relax the computation cost of (46).

As an alternative to KCEM, there are also two recent approaches to generalize CEM to nonlinear versions of CEM.

2) Hierarchical CEM: HCEM implements a series of concatenated CEM detectors where each concatenated CEM detector takes the output of its preceding CEM detector and processes it by a nonlinear function, which is very similar to kernels used in multilayer neural networks and also a nonlinear function used in KCEM specified by (43). Then, the output of the nonlinear filter will be further used as the input to its follow-up cascaded CEM detector. Two parameters are required to implement HCEM. One is the nonlinear function, which is generally a sigmoid function specified by a parameter λ. The other is a threshold, which determines the number of concatenated CEM detectors, k.

3) Ensemble Cascaded CEM: ECEM is specifically designed to improve the detection of CEM over nonlinearity and generalization ability. It consists of two-stage processes, multiscale scanning process, and a cascade of CEM detectors. The purpose of the multiscale scanning process is to use a sliding window to scan and crop the entire spectrum into a set of spectral segments, each of which will be further processed by a CEM detector to produce its feature vectors. These CEM-obtained feature vectors are further concatenated to produce a final feature vector as an input to the follow-up a cascade of CEM detectors in the second-stage process where each cascaded CEM detector is implemented by a random regularization parameter to enhance CEM's generalizability and further processed by a sigmoid function to make a CEM detector a nonlinear filter. There are several parameters that need to be tuned to implement ECEM, which are the number of sliding windows n, the number of cascaded layers k, and the number of CEM detectors used in each cascaded layer m.

B. Iterative NCEM

In hyperspectral imaging, CEM was originally designed for subpixel target detection, which is solely operated based on a single-pixel vector without accounting for interpixel correlation. As a result, it is very effective when it is used to detect targets at a subpixel scale. However, on many occasions, a target to be detected may occupy more than one single-pixel vector or even be mixed with BKG. Under such a circumstance, interpixel correlation may be needed to be factored in to help to improve target detection. Recently, an ICEM was developed for HSIC [11] where CEM was used as a subpixel classifier for each of classes of interest to be classified. As noted above, CEM is effective as a subpixel target detector and does not take care of interpixel spatial information. To mitigate this dilemma, ICEM introduces an iterative feedback process that implements a set of spatial filters to capture spatial information from CEM-detection maps and feeds the obtained spatial-filtered classification information back and added to the current image data cube to create a new image data cube for next round iteration by re-implementing CEM. Despite the fact that CEM is still linear, the spatial filters used to capture the spatial information are generally nonlinear such as Gaussian, edge-preserving filters, and Gabor filters. In this case, ICEM can be considered as a linear filter operating nonlinear expanded data cubes in an iterative process. Now, if we further replace CEM in ICEM with NCEM to yield INCEM, this nonlinear spatial-filtered information works well in a similar manner that an NCEM uses a nonlinear function to capture spatial information. Interestingly, while INCEM has been widely used in HSIC [19], its application to target detection has yet to be investigated. This section presents an iterative version of NCEM, also referred to as INCEM for target detection.

INCEM for Target Detection

1. Initial condition: d is given. $k = 0$, $\mathbf{\Delta}^{(0)} = \mathrm{HSI}$ is the original image data cube, HSI.
2. At the kth iteration, operating $\delta^{\mathrm{NCEM}}(\mathbf{r})$ on $\mathbf{\Delta}^{(k)}$ to produce an NCEM-detection map $\mathrm{NCEMap}^{(k)}$.
3. Apply Otsu's method to $\mathrm{NCEMap}^{(k)}$ to produce a binary map $\mathrm{BNCEMap}^{(k)}$.
4. If $k = 0$, go to step 2. Otherwise, continue.
5. Check if the Tanomoto index (TI) [31] defined by

$$\mathrm{TI}^{(k)} = \frac{\left| \mathrm{BNCEMap}^{(k-1)} \cap \mathrm{BNCEMap}^{(k)} \right|}{\left| \mathrm{BNCEMap}^{(k-1)} \cup \mathrm{BNCEMap}^{(k)} \right|} > \varepsilon \tag{47}$$

   where $|A|$ is the size of a set A. If it is not, let $k \leftarrow k + 1$ and go to step 2. Otherwise, continue.
6. If (36) it is not satisfied, form a new expanded image data cube by

$$\mathbf{\Delta}^{(k)} = \mathbf{\Delta}^{(k-1)} \cup \mathrm{NCEMap}^{(k)}. \tag{48}$$

   Let $k \leftarrow k + 1$ and go to step 2. Otherwise, continue.
7. Output $\mathrm{NCEMap}^{(k)}$ for final target detection.

It is worth noting that the NCEM implemented in INCEM can be any nonlinear version of CEM such as KCEM, HCEM, and ECEM to yield IKCEM, IHCEM, and IECEM so as to improve the performance of ICEM.

C. Iterative RTS NCEM

Upon implementing CEM, the prior target knowledge $\mathbf{d}$ must be known or provided in advance. Generally, there are two scenarios. One is provided by GT. The other is provided by training samples. In practical applications, obtaining the complete GT may not be feasible. Therefore, an approach to resolving this issue is to use training samples to generate the desired knowledge of $\mathbf{d}$ to replace the GT-provided target knowledge. In other words, using training samples may be more realistic than GT. In doing so, the IRTS-CEM developed for HSIC in [14] is modified to IRTS-NCEM for target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

detection by implementing INCEM described in Section IV-B coupled with a concept of RTS [13] to make IRTS-NCEM a random detector via an iterative feedback process similar to ICEM. More precisely, I-NCEM uses GT to calculate $\mathbf{d}$, whereas IRTS-NCEM is designed to use RTS to calculate $\mathbf{d}$. Specifically, at each iteration executed by IRTS-NCEM, the training samples are randomly reselected from the expanded image data cube, which is generated by adding the spatial-filtered NCEM-detection maps obtained at the preceding iteration to the currently being processed image data cube in exactly the same way as INCEM does. However, IRTS-NCEM includes RTS as an additional process to replace the complete GT-calculated $\mathbf{d}$ with RTS-calculated $\mathbf{d}$. In this case, RTS makes the RTS at each iteration dynamic so that each labeled data sample will be equally accessed by RTS at each iteration through the entire iterative process. This implies that if a labeled data sample was not selected at one iteration, it may be likely to be selected in subsequent iterations. Therefore, the more iterations are carried out, the more labeled data samples will be selected to update $\mathbf{d}$.

IRTS-NCEM for Target Detection

1. Initial condition: $k = 0$, $\mathbf{\Delta}^{(0)} = \text{HSI}$ is the original image data cube, HSI and $S^{(0)}$ is an initial randomly selected training samples which are used to find the initial target signature, $\mathbf{d}^{(0)}$.
2. At the kth iteration, randomly select a set of training samples from $\mathbf{\Delta}^{(k)}$, denoted by $S^{(k)}$ which is used to find the desired target signature $\mathbf{d}^{(k)}$.
3. Operate $\delta^{\text{NCEM}}(\mathbf{r})$ on $\mathbf{\Delta}^{(k-1)}$ using $\mathbf{d}^{(k)}$ as the kth desired target signature to produce an NCEM-detection map $\text{NCEMap}^{(k)}$.
4. Apply Otsu's method to $\text{NCEMap}^{(k)}$ to produce a binary map $\text{BNCEMap}^{(k)}$.
5. If $k = 0$, go to step 2. Otherwise, continue.
6. Check if TI defined by (47).
7. If (36) is not satisfied, a new expanded image data cube is created by (37). Let $k \leftarrow k + 1$ and go to step 2. Otherwise, continue.
8. Output $\text{NCEMap}^{(k)}$ for final target detection.

Since IRTS-NCEM includes INCEM as part of its process, KCEM, HCEM, and ECEM can also be used to replace NCEM in IRTS-NCEM to yield IRTS-KCEM, IRTS-HCEM, and IRTS-ECEM.

D. Multiple-Target CEM

CEM was designed as a single-target detection. If there is more than one target, CEM must be implemented multiple times. This section generalizes CEM to multitarget CEM, which can detect multiple targets simultaneously at the same time.

1) Linearly Constrained MV: In CEM, the MVDR filter used a single constraint to constrain a desired target signature $\mathbf{d}$. In [20], Frost extended CEM to LCMV filter, which allows multiple linear constraints to adjust an array of sensors in real time to respond to a set of desired signals specified by $[\mathbf{D}]$ while discriminating against noise. By using multiple constraints on weights of $\mathbf{D}$ in the array, the LCMV filter can be shown to detect multiple targets with no need of prior knowledge of undesired signals or noise statistics. The resulting LCMV filter is called LCMV target detector.

Assume that $\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_M$ are M specific target signatures of interest, which can be either provided by a priori knowledge or obtained by training samples and so on. Suppose that $\{\mathbf{r}_i\}_{i=1}^N$ is the set of data sample vectors in a hyperspectral image, where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$ is the ith L-dimensional data sample vector and L is the total number of spectral bands.

Now, we interpret each of $\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_M$ as a desired signal arrival direction in adaptive beamforming and let $\mathbf{D}^M = [\mathbf{d}_1\ \mathbf{d}_2\ \cdots\ \mathbf{d}_M]$ be an $L \times M$ target signature matrix, where L is the total number of band. Assume that an FIR linear filter is specified by L filter coefficients $L \times M$, denoted by an L-dimensional vector $L \times M$. The LCMV problem can be then reformulated as the following M-target-constrained optimization problem:

$$\min_{\mathbf{w}}\left\{\mathbf{w}^T\mathbf{R}\mathbf{w}\right\} \quad \text{s.t. } \left(\mathbf{D}^M\right)^T\mathbf{w} = \mathbf{c}^M \tag{49}$$

where $\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\}$ in (49) was referred to as MV, $\mathbf{c}^M = (c_1, c_2, \ldots, c_M)^T$ is an M-dimensional constraint vector for a general purpose, and $\mathbf{R} = (1/N)\sum_{i=1}^N \mathbf{r}_i\mathbf{r}_i^T$ is the sample correlation matrix of size $L \times L$. It should be noted that each of M constraints, $c_1, c_2, \ldots, c_M$, in (37) is used to impose on a particular target, i.e., the mth constraint, c_m is imposed on the mth target $\mathbf{d}_m$ and chosen to be any arbitrary M-dimensional vector. For example, in adaptive beamforming of array signal processing, $\mathbf{c}^M$ is only used to lock signal arrivals in M desired directions. For simplicity, we can choose $\mathbf{c}^M$ to be the M-dimensional unity vector with ones in all the M components, $\mathbf{1}^M = (1, \ldots, \underbrace{1}_{j}, \ldots, \underbrace{1}_{M})^T$. In LCMV, these M signal arrival directions are specified by M target signatures. Therefore, we can also use the M-dimensional unity vector to constrain the target signature matrix $\mathbf{D}^M$. In this case, the optimal solution to (47) with $\mathbf{c}^M$ replaced by $\mathbf{1}^M$ is solved by

$$\mathbf{w}^* = \mathbf{R}^{-1}\mathbf{D}^M\left(\left(\mathbf{D}^M\right)^T\mathbf{R}^{-1}\mathbf{D}^M\right)^{-1}\mathbf{1}^M \tag{50}$$

where $(\mathbf{D}^M)^T\mathbf{R}^{-1}\mathbf{D}^M$ is an $M \times M$ matrix.

Then, LCMV is specified by

$$\delta^{\text{LCMV}}(\mathbf{r}) = \left(\mathbf{w}^*\right)^T\mathbf{r} \tag{51}$$

which performs multitarget detection with M target signature matrix $\mathbf{D}^M$ constrained by $\mathbf{1}^M$ and BKG suppressed by $\mathbf{R}^{-1}$.

Now, substituting (48) into (49) yields

$$\left(\mathbf{w}^*\right)^T\mathbf{R}\mathbf{w}^* = \left(\mathbf{1}^M\right)^T\left(\left(\mathbf{D}^M\right)^T\mathbf{R}^{-1}\mathbf{D}^M\right)^{-1}\mathbf{1}^M \tag{52}$$

which is the MV resulting from LCMV in (50).

2) Target-Constrained Interference Minimization Filter: On many practical applications, some partial knowledge on the BKG or unwanted signal sources may be available. In this case, we should take advantage of such knowledge and use it as undesired signal knowledge $[\mathbf{U}]$ for elimination prior to detection so as to improve target detection performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

In doing so, LCMV can be combined with OSP to derive a TCIMF developed by Ren and Chang [21] where the constraint imposed on desired multiple-target signature set [**D**] in LCMV is expanded to also include undesired target signature set [**U**] to form a new constrained multiple-target signature [**D**, **U**]. Therefore, to fully implement TCIMF, we should explore the multiple-target signatures to be constrained [**D**, **U**].

Let $\{\mathbf{r}_i\}_{i=1}^N$ be the set of data sample vectors in a hyperspectral image, where $\mathbf{r}_i = (r_{i1}, r_{i2}, \ldots, r_{iL})^T$, $1 \le i \le N$, represents an L-dimensional sample vector and N and L are the total number of pixels and spectral bands, respectively. In addition, let $\mathbf{D} = [\mathbf{d}_1\ \mathbf{d}_2\ \cdots\ \mathbf{d}_p]$ be an $(L \times p)$-dimensional signature matrix of interest and denote $\mathbf{U} = [\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_q]$ as an $(L \times q)$-dimensional undesired target signature matrix in detection. It should be noted that **D** and **U** can be either provided by a priori knowledge or obtained by a posteriori knowledge, respectively. TCIMF is designed as a target detector that can detect signals specified by the desired target signature matrix **D** while annihilating undesired target signature matrix **U** via an FIR filter specified by an L-dimensional vector $\mathbf{w} = (w_1, w_2, \ldots, w_L)^T$. Specifically, the TCIMF problem can be formulated as the following constrained formulation:

$$\min_{\mathbf{w}}\{\mathbf{w}^T\mathbf{R}\mathbf{w}\} \quad \text{s.t. } [\mathbf{D}\ \mathbf{U}]^T\mathbf{w} = \begin{bmatrix}\mathbf{1}_{p\times 1}\\ \mathbf{0}_{q\times 1}\end{bmatrix} \tag{53}$$

where $[\mathbf{D}\ \mathbf{U}]$ is defined as $[\mathbf{D}\ \mathbf{U}] = [\mathbf{d}_1\ \mathbf{d}_2\ \cdots\ \mathbf{d}_p\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_q]$, $\mathbf{1}_{p\times 1}$ is a $p \times 1$ vector with ones in all components, and $\mathbf{0}_{q\times 1}$ is a $q \times 1$ vector with zero in its components. In order to constrain the desired target and undesired target signatures, the $(p \times q)$-dimensional vector $\mathbf{c} = (c_1, c_2, \ldots, c_p, \ldots, c_q)^T$ is used and specified by

$$\mathbf{c} = \begin{bmatrix}\mathbf{1}_{p\times 1}\\ \mathbf{0}_{q\times 1}\end{bmatrix} = \left(\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}\right)^T. \tag{54}$$

Then, the optimal weight vector $\mathbf{w}^{\mathrm{TCIMF}}$ solved for (50) is given by

$$\mathbf{w}^{\mathrm{TCIMF}} = \mathbf{R}^{-1}[\mathbf{D}\ \mathbf{U}]\left([\mathbf{D}\ \mathbf{U}]^T\mathbf{R}^{-1}[\mathbf{D}\ \mathbf{U}]\right)^{-1}\begin{bmatrix}\mathbf{1}_{p\times 1}\\ \mathbf{0}_{q\times 1}\end{bmatrix} \tag{55}$$

which yields

$$\delta^{\mathrm{TCIMF}}(\mathbf{r}) = \left(\mathbf{R}^{-1}[\mathbf{D}\ \mathbf{U}]\left([\mathbf{D}\ \mathbf{U}]^T\mathbf{R}^{-1}[\mathbf{D}\ \mathbf{U}]\right)^{-1}\begin{bmatrix}\mathbf{1}_{p\times 1}\\ \mathbf{0}_{q\times 1}\end{bmatrix}\right)^T\mathbf{r}. \tag{56}$$

When $\mathbf{U} = \varnothing$, $\delta^{\mathrm{TCIMF}}(\mathbf{r})$ in (56) is reduced to LCMV target detector, $\delta^{\mathrm{LCMV}}(\mathbf{r})$ in (51).

Compared to LCMV in (51) and OSP in (1), TCIMF combines the strengths of both of them, by suppressing BKG by $\mathbf{R}^{-1}$ as LCMV does and annihilating the impact of undesired target signatures in **U** as OSP does by (2). Specifically, TCIMF expands constraint vector **1** to $[\mathbf{1}_{p\times 1}\ \mathbf{0}_{q\times 1}]^T$, where component "**1**" is used to constrain the particular desired target signatures, while component "**0**" is added to annihilate the undesired target signatures. Because of these two joint constraints imposed on the desired targets and undesired targets, TCIMF can improve the detection ability of the detector on the targets of interest as desired targets by imposing the constraint vector "**1**" as well as the constraint vector "**0**" on undesired targets to eliminate the influence of undesired targets on the desired targets. In the meantime, TCIMF can also minimize the interfering effects resulting from BKG and noise by inverting the sample correlation matrix **R**.

3) Kernel Target-Constrained Interference-Minimized Filter: Following the same treatment, which drives (42)–(44), a kernel version of TCIMF can also be derived as follows:

$$\mathbf{w}_{\varphi}^{\mathrm{TCIMF}} = \mathbf{R}_{\varphi}^{-1}\varphi([\mathbf{D}\ \mathbf{U}])\left(\varphi([\mathbf{D}\ \mathbf{U}])^T\mathbf{R}_{\varphi}^{-1}\varphi([\mathbf{D}\ \mathbf{U}])\right)^{-1}\begin{bmatrix}\mathbf{1}_{p\times 1}\\ \mathbf{0}_{q\times 1}\end{bmatrix}_{\varphi} \tag{57}$$

where $\mathbf{R}_{\varphi}$ is defined by (42)

$$\begin{aligned}\delta^{\mathrm{KTCIMF}}(\mathbf{r}_n) &= \left(\mathbf{w}^{\mathrm{KTCIMF}}\right)^T\varphi(\mathbf{r}_n)\\ &= \frac{\varphi\left([\mathbf{D}\ \mathbf{U}]^T\right)\mathbf{R}_{\varphi}^{-1}\varphi(\mathbf{r}_n)}{\varphi\left([\mathbf{D}\ \mathbf{U}]^T\right)\mathbf{R}_{\varphi}^{-1}\varphi([\mathbf{D}\ \mathbf{U}])}\mathbf{c}_{\varphi}\end{aligned} \tag{58}$$

where

$$\mathbf{w}^{\mathrm{KTCIMF}} = \frac{\mathbf{R}_{\varphi}^{-1}\varphi([\mathbf{D}\ \mathbf{U}])}{\varphi\left([\mathbf{D}\ \mathbf{U}]^T\right)\mathbf{R}_{\varphi}^{-1}\varphi([\mathbf{D}\ \mathbf{U}])}\mathbf{c}_{\varphi}. \tag{59}$$

Like (46), it is impossible to solve (58) due to the infinite dimensionality in feature space H. Therefore, an RBF kernel function defined in (45) is also used to calculate (58). However, computing the kernel matrix **K** in (46) is very expensive in terms of computational times and memories. In this case, low-rank approximation, such as Nyström method [19], is used to relax the computation cost of (58). Specifically, when $\mathbf{U} = \varnothing$, $\delta^{\mathrm{KTCIMF}}(\mathbf{r})$ in (58) is reduced to KLCMV target detector. On the other hand, if $[\mathbf{DU}] = \mathbf{d}$, then $\delta^{\mathrm{KTCIMF}}(\mathbf{r})$ in (58) is reduced to KCEM target detector in (46).

4) BKG-Annihilated TCIMF: Although KTCIMF can suppress unknown BKG via inverting the sample correlation matrix **R**, the BKG of a hyperspectral scene can be very complicated due to the fact that many unknown or unidentified signal sources can be uncovered by a hyperspectral imaging sensor. To better cope with BKG, a BA-KTCIMF was recently developed by Chen and Chang [25], which suppresses BKG in two stages. In the first stage, the global BKG is suppressed by three approaches, DS [32, Ch. 6], decomposition of a data space **X** into a low-rank subspace **L**, and sparse subspace **S** such as low-rank and sparse matrix decomposition (LRaSMD) proposed in [33] and CDA in [34], which decomposed a data space into principal component (PC), independent component (IC), and noise component. In the second stage, additional BKG signatures are extracted from the BKG-annihilated data as unwanted signatures so that TCIMF can further enhance TD and suppress target BKG in the BKG-annihilated data.

Depending upon how these two stages are carried out, three versions of BA-KTCIMF, data sphered BA-KTCIMF (DS-BA-KTCIMF), LRaSMD BA-KTCIMF (LRaSMD-BA-KTCIMF), and CDA-BA-KTCIMF, are derived. Fig. 3 depicts a general block diagram of two stages implemented in BA-TCIMF where TCIMF is used as a generic term that can be replaced by various versions of TCIMF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

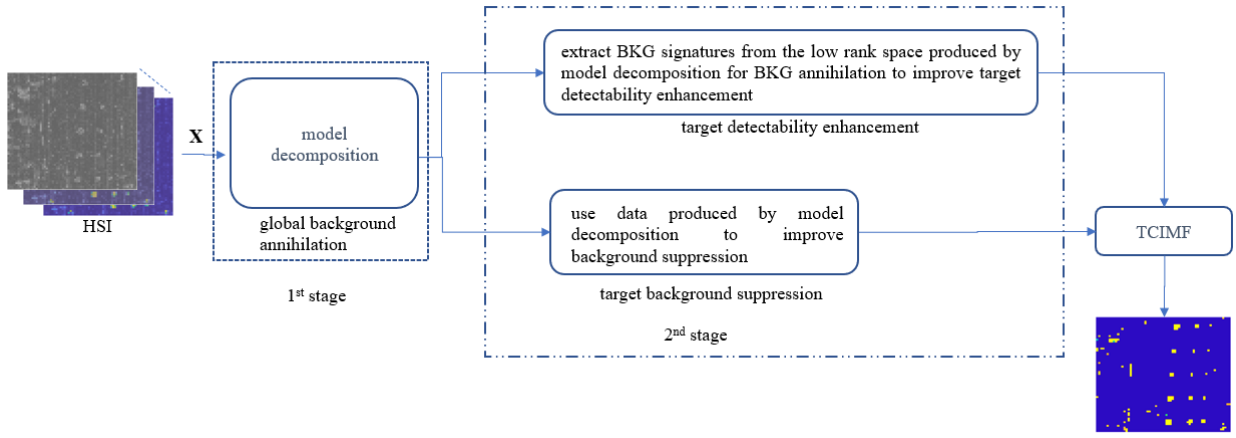


Fig. 3. Two stages implemented in BA-TCIMF.

VI. Three-Dimensional ROC Curve-Derived Detection Measures

To conduct a detailed quantitative analysis on detection performance, a recent work in [35] proposed an effective 3-D ROC analysis-based evaluation tool, which includes the following several 3-D ROC curve-derived detection measures that were used for performance evaluation.

Assume that a 3-D ROC is generated by a triplet of parameters (P_D, P_F, τ). Then, a 3-D ROC curve can derive three 2-D ROC curves of (P_D, P_F), (P_D, τ), and (P_F, τ). Let AUC be the area under these 2-D ROC curves denoted by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$. By virtue of the values of $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$, we can define the following quantitative detection measures.

1) TD is denoted by AUC_{TD} and defined by
$$0 \le AUC_{TD} = AUC_{(D,F)} + AUC_{(D,\tau)} \le 2. \quad (60)$$
2) BS measure is denoted by AUC_{BS} and defined by
$$-1 \le AUC_{BS} = AUC_{(D,F)} - AUC_{(F,\tau)} \le 1. \quad (61)$$
3) Target detection and BS (TDBS) is denoted by AUC_{TDBS} and defined by
$$-1 \le AUC_{TDBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 1. \quad (62)$$
4) ODP is denoted by AUC_{ODP} and defined by
$$-1 \le AUC_{ODP} = AUC_{(D,F)} + AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 2. \quad (63)$$
5) SNPR is denoted by AUC_{SNPR} and defined by
$$0 \le AUC_{SNPR} = \frac{AUC_{(D,\tau)}}{AUC_{(F,\tau)}}. \quad (64)$$

The above detection measures can be further categorized into four different functionalities.

1) *TD:* $AUC_{(D,\tau)}$ and AUC_{TD}.
2) *BKG Suppressibility:* $AUC_{(F,\tau)}$ and AUC_{BS}.
3) *Joint TD and BKG Suppressibility:* AUC_{TDBS} and AUC_{SNPR}.
4) *Detector's Effectiveness:* $AUC_{(D,F)}$ and AUC_{ODP}.

These eight detection measures were then used to evaluate the quantitative performance of detectors for experiments conducted in the following section.

VII. Image to Be Studied for Experiments

There are many real datasets that can be used for HTD. It is impossible to run experiments for most of the datasets. Since the theory and generalization of CEM is the main goal of this article, in this section, we particularly selected two datasets to validate these claims. For more discussions on an issue of how to select datasets for experiments, we refer details to [36].

A. HYDICE Panel Data

An airborne hyperspectral digital imagery collection experiment (HYDICE) scene shown in Fig. 4(a) was used to analyze comparative performance among various target detectors and anomaly detectors. It was collected in August 1995 from a flight altitude of 10 000 ft. Here are 15 square panels in Fig. 4(a) with three different sizes, 3 × 3, 2 × 2, and 1 × 1 m. Due to the ground sampling distance of approximately 1.56 m, each of the panels in the first column except the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 1. All the remaining 11 panels in Fig. 4 contain one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 4(b) shows their precise spatial locations with the pixels in yellow (Y pixels) indicating panel pixels mixed with the BKG. Fig. 4(c) plots the five panel spectral signatures, $\mathbf{p}_1$–$\mathbf{p}_5$ obtained from Fig. 4(b), where the ith panel signature denoted by $\mathbf{p}_i$ was generated by averaging the red panel center pixels in row i. These five panel spectral signatures were used as the desired target signature for CEM. Fig. 4(c) plots the five panel spectral signatures obtained from Fig. 4(b), where the ith panel signature denoted by $\mathbf{p}_i$ was generated by averaging the red panel center pixels in row i. These panel signatures will be used to represent the target knowledge of the panels in each row.

The HYDICE image scene in Fig. 4 has three unique advantages. One is the GT provided by Fig. 4(b) to show the exact spatial locations of each of the 19 R panel pixels shown in Fig. 4(b) so that each detector can be evaluated by detection of 19 R panel pixels up to pixel level. Another is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

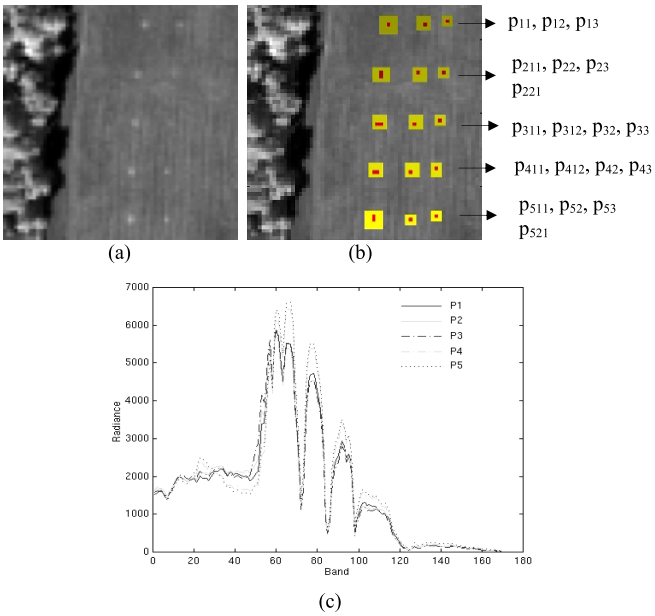


Fig. 4. (a) HYDICE panel scene which contains 15 panels. (b) GT map of spatial locations of 19 R panel pixels. (c) Spectra of $\mathbf{p}_1$, $\mathbf{p}_2$, $\mathbf{p}_3$, $\mathbf{p}_4$, and $\mathbf{p}_5$.

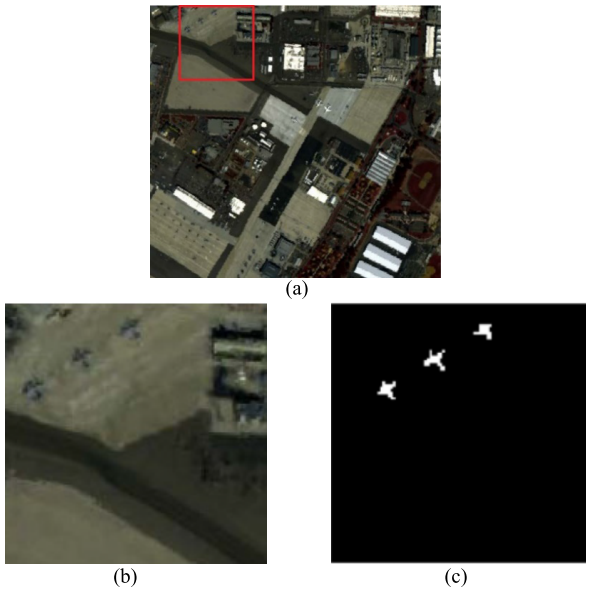


Fig. 5. AVIRIS San Diego airport scene. (a) Pseudo-color image of the whole scene. (b) Pseud-color image of the selected area. (c) GT map.

the knowledge of desired target signatures that is only up to three or four pixels in each row by GT, and there are no sufficient labeled data samples that can be used as training samples learning. As a consequence, deep learning (DL) based methods are not applicable to this scene to justify their learning effectiveness. A third one is to evaluate the ability of a target detector to detect pure pixels in the first column, mixed pixels in the second column, and subpixels in the third column.

There are three reasons why the HYDICE panel data was selected for experiments.

1) The HYDICE panel data have five subpixel panels, p_{13}, p_{23}, p_{33}, p_{43}, and p_{53}, located in the third column of the scene, which cannot be visualized by the human naked eye because their size of 1 × 1 m is smaller than the image spatial resolution, 1.56 m. In this case, subpixel detectability of CEM can be validated by finding these five subpixel panels.
2) To the author's best knowledge, this panel dataset is probably only real dataset that has a complete GT of subpixel targets that can be used to validate subpixel detection algorithms.
3) Since there are so few target data samples, i.e., only three panel samples for row 1 and four panel samples for rows 2–5, training-based DL methods may not be applicable due to a lack of training samples to be used for training.

B. San Diego Airport Data

A second dataset shown in Fig. 5(a) is an AVIRIS airport scene from San Diego, CA, USA. The size of the airport is 400 × 400 pixels, and its spatial resolution is 3.5 m with 224 spectral bands, which is the third dimensionality ranging from 370 to 2510 nm. Some bands corresponding to the water absorption regions, the low signal-to-noise ratio, and collapse bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224) have been removed and finally result in 189 bands. A red square marked at the upper left corner of the scene was chosen as the test image with its ground-truth image shown in Fig. 5(b) and (c). The restricted area has a size of 100 × 100 pixels. the BKG of the scene is mainly made up of a roof, shadow, and grass. The targets of interest in this image scene are three airplanes with the same materials in the scene, which comprised 85 pixels and account for 0.33% of the test image, as shown in Fig. 5(b).

Since the locations of these planes can be obtained from GT in Fig. 5(c), the extracted 85 target pixels from these three airplanes can be used as labeled data samples to calculate the desired signature $\mathbf{d}$. Unlike the HYDICE panel data in Fig. 4, which has only 3–4 pixels for each row, the airplanes in Fig. 5 have a sufficient number of labeled data samples that can be used as training samples.

This dataset has three airplanes that are very large and visible. In this case, training samples can be selected by visual inspection. It has been one of the widely used datasets in the literature for both target detection and anomaly detection. There are also two reasons why San Diego airport data were selected for experiments.

1) Unlike the HYDICE panel data, this dataset has a large number of target data samples that can be used as training samples. This explains why this dataset is so popular to be used for DL-based methods. The purpose of using this dataset was to demonstrate that CEM not only can detect subpixel panels such as subpixel targets in HYDICE data with no need of training samples but also can detect large targets by using training samples without appealing for DL methods.
2) This dataset does not have subpixel targets of interest. As a result, many DL-based methods, which claimed to perform well on this dataset, may not do well for subpixel targets. For example, please see [43, Fig. 18].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

VIII. Analysis of Experimental Results

According to [3], CEM, HCEM, and ECEM were shown to perform better than many existing target detectors, such as adaptive cosine estimate (ACE), adaptive matched filter (AMF), and adaptive subspace detector (ASD). In this case, CEM, HCEM, and ECEM were used as benchmark target detectors for comparison to conduct a comprehensive study and analysis.

A. HYDICE Panel Data

Fig. 6 shows the detection maps of 19 R panels in Fig. 4 produced by 17 target detectors: (a) CEM, (b) ICEM, (c) KCEM, (d) IKCEM, (e) LCMV, (f) ILCMV, (g) TCIMF, (h) ITCIMF, (i) KTCIMF, (j) IKTCIMF, (k) HCEM, (l) IHCEM, (m) ECEM, (n) IECEM, (o) DS-BA-KTCIMF, (p) LRaSMD-BA-KTCIMF, and (q) CDA-BA-KTCIMF. By visual examination of Fig. 6, the best results seemed to be those produced by (d) IKCEM, (k) HCEM, (l) IHCEM, (o) DS-BA-KTCIMF, (p) LRaSMD-BA-KTCIMF, and (q) CDA-BA-KTCIMF where the exact locations of 19 R pixels were detected correctly with very clean BS. Interestingly, (g) TCIMF, (h) ITCIMF, (i) KTCIMF, and (j) IKTCIMF did not perform as well as IKCEM did. This is because the panel signature $\mathbf{p}_2$ in row 2 and $\mathbf{p}_3$ in row 3 are very similar. Therefore, when detecting $\mathbf{p}_2$ with $\mathbf{p}_3$ in $\mathbf{U}$, the performance of TCIMF in detecting $\mathbf{p}_2$ was degraded by removing $\mathbf{p}_3$ for annihilation as can be seen from Fig. 6(g)–(j), specifically panels in row 3 where IKTCIMF did very well except the subpixel panel p_{33} in the third column. However, this issue was corrected by (o) DS-BA-KTCIMF, (p) LRaSMD-BA-KTCIMF, and (q) CDA-BA-KTCIMF by including target BS operated in the second stage shown in Fig. 3.

The results in Fig. 6 are qualitative. To further conduct quantitative analysis, the eight detection measures derived in Section VI were calculated and tabulated in Tables I–V for the detection of panels in five different rows where the best results are boldfaced. As shown in Tables I–V, IHCEM was the best because it produced all the eight best values across board except Table I, which produced seven best values except $\mathrm{AUC}_{(\mathrm{D},\tau)}$. Nonetheless, the three versions of BA-KTCIMF also did very well with differences in the range of 10^{-3} and can be considered as the second-best target detectors.

B. AVIRIS San Diego Airport Scene

Since the airplanes in Fig. 5 are the only target of interest, $\mathbf{D} = \mathbf{d}$. In this case, the multiple-target CEM, LCMV, is reduced to CEM. In addition, there is no prior knowledge about $\mathbf{U}$, i.e., $\mathbf{U} = \varnothing$ in which case TCIMF is also reduced to CEM as well. Therefore, the target detectors used for experiments were CEM, ICEM, KCEM, HCEM, IHCEM, ECEM, and IECEM.

1) $\mathbf{d}$ *Calculated by GT:* Fig. 7 shows their detection maps where the desired target signature $\mathbf{d}$ was calculated by averaging the 85 target pixels by GT. By visual inspection, the best results seemed to be those obtained by IKCEM and IHCEM with very clean BS. However, it is very difficult to determine which one was the best one. To resolve this issue, the eight detection measures derived in Section VI were used for quantitative analysis. Table VI calculated the AUC values obtained by eight detection measures from Fig. 7 where the best results were boldfaced. According to Table VI, IKCEM and IHCEM were the best target detectors as confirmed in Fig. 7, each of which produced the four best results. Specifically, IKCEM was the best target detector with the best TD by producing the best values of $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$ compared to IHCEM, which was the best target detector with the best BS by producing the best values of $\mathrm{AUC}_{(\mathrm{D},\mathrm{F})}$, $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$.

2) $\mathbf{d}$ *Calculated by Training Samples:* Unlike Section VIII-B1 where $\mathbf{d}$ was calculated by GT, this section performed experiments with $\mathbf{d}$ calculated by training samples. In this case, ICEM, IKCEM, IHCEM, and IECEM were replaced by IRTS-CEM, IRTS-KCEM, IRTS-HCEM, and IRTS-ECEM, respectively, where training samples were randomly selected 10%, 20%, 30%, 40%, and 50% from the total number of the GT-ed 85 target pixels to calculate $\mathbf{d}$.

Fig. 8(a)–(e) shows the detection maps of IRTS-CEM. By visual inspection, all the results are comparable with only a slight difference in BS.

To conduct quantitative analysis, Table VII calculated the AUC values by eight detection measures from Fig. 8 where the best results were boldfaced. Interestingly, IRTS-CEM using 30% was the best target detector with the best TD by producing the best values of $\mathrm{AUC}_{(\mathrm{D},\tau)}$, $\mathrm{AUC}_{\mathrm{TD}}$, $\mathrm{AUC}_{\mathrm{TDBS}}$, and $\mathrm{AUC}_{\mathrm{ODP}}$ compared to IRTS-CEM, which was the best target detector with the best BS by producing the best values of $\mathrm{AUC}_{(\mathrm{D},\mathrm{F})}$, $\mathrm{AUC}_{(\mathrm{F},\tau)}$, $\mathrm{AUC}_{\mathrm{BS}}$, and $\mathrm{AUC}_{\mathrm{SNPR}}$.

Fig. 9(a)–(e) shows the detection maps of IRTS-KCEM. Compared to Fig. 8(a)–(e), KCEM had much greater BS than IRTS-CEM and clearly cleaned up most BKG and also pixels mixed with the three airplanes. It should be noted that the brightest pixels on the three airplanes were those pixels particularly specified by the training samples calculated $\mathbf{d}$. To further conduct quantitative analysis, Table VIII calculated the AUC values by eight detection measures from Fig. 9 where the best results were boldfaced where all the AUC values increased monotonically as the training sample rate was increased from 10% to 50% except the 40% case that its $\mathrm{AUC}_{(\mathrm{F},\tau)}$ and $\mathrm{AUC}_{\mathrm{SNPR}}$ values yielded the best values. Nevertheless, by visual inspection of Fig. 9(c)–(e), their detection maps are nearly the same and have no appreciable difference.

Fig. 10(a)–(d) shows the detection maps of IRTS-HCEM. Compared to Fig. 9(a)–(d), HCEM performed comparably to KCEM due to the fact they both used kernel functions. Interestingly, for the case of using 50% training sample rate, IRTS-HCEM did much better than IRTS-KCEM in terms of detecting three airplanes. To further conduct quantitative analysis, Table IX calculated the AUC values by eight detection measures from Fig. 10 where the best results were boldfaced where all the AUC values increased monotonically as the training sample rate was increased from 10% to 50% except the 10% case that its $\mathrm{AUC}_{(\mathrm{F},\tau)}$ yielded the best value. It should be noted the results of using 50% training sample rate performed considerably well compared to the results of using other smaller training sample rates.

Interestingly, by visually comparing the detection map produced by IRTS-HCEM in Fig. 9(e) to the detection map

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

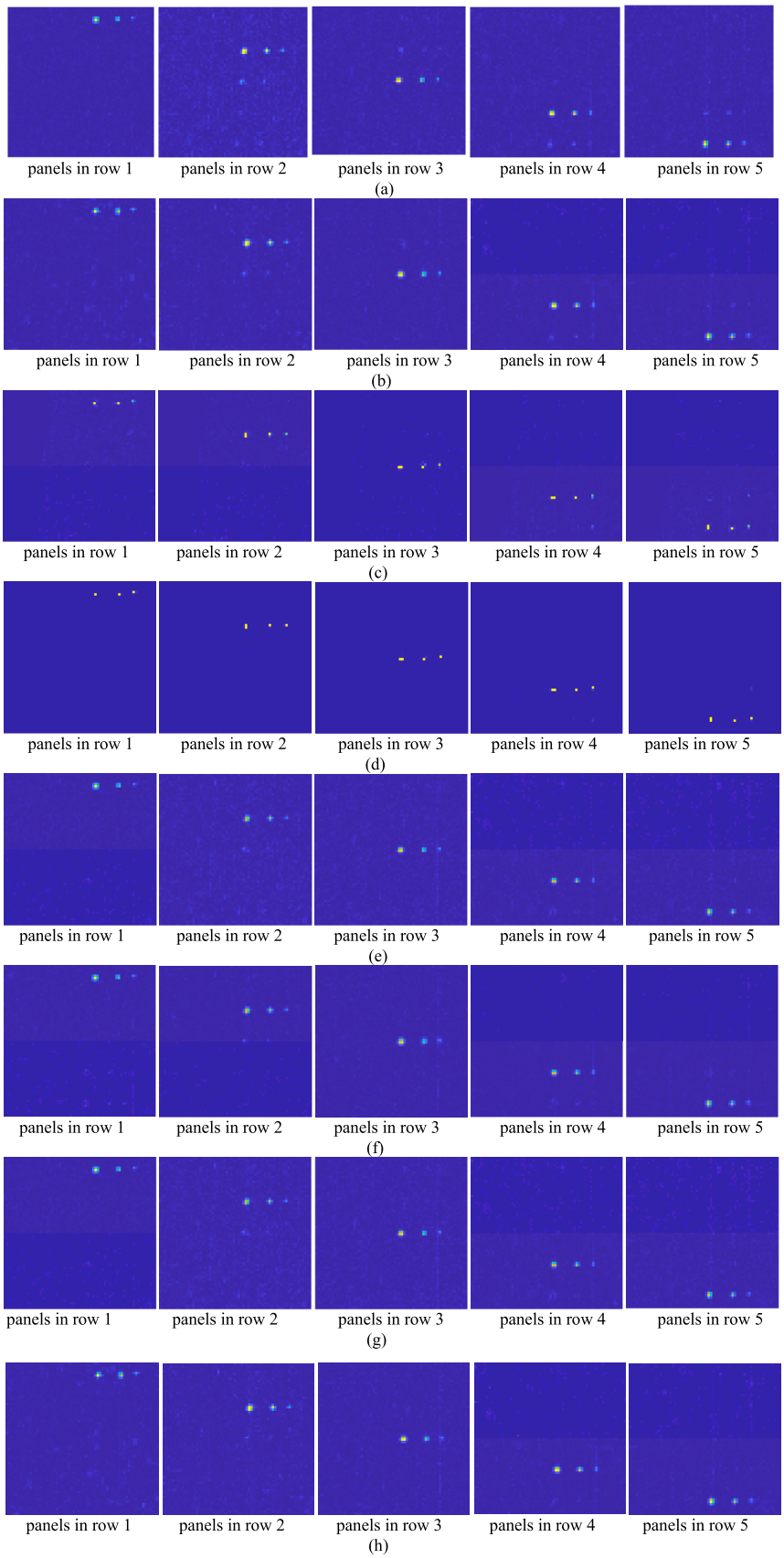


Fig. 6. Target abundance detection maps of various target detectors for HYDICE data scene. Detection maps of (a) CEM, (b) ICEM, (c) KCEM, (d) IKCEM, (e) LCMV, (f) ILCMV, (g) TCIMF, and (h) ITCIMF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

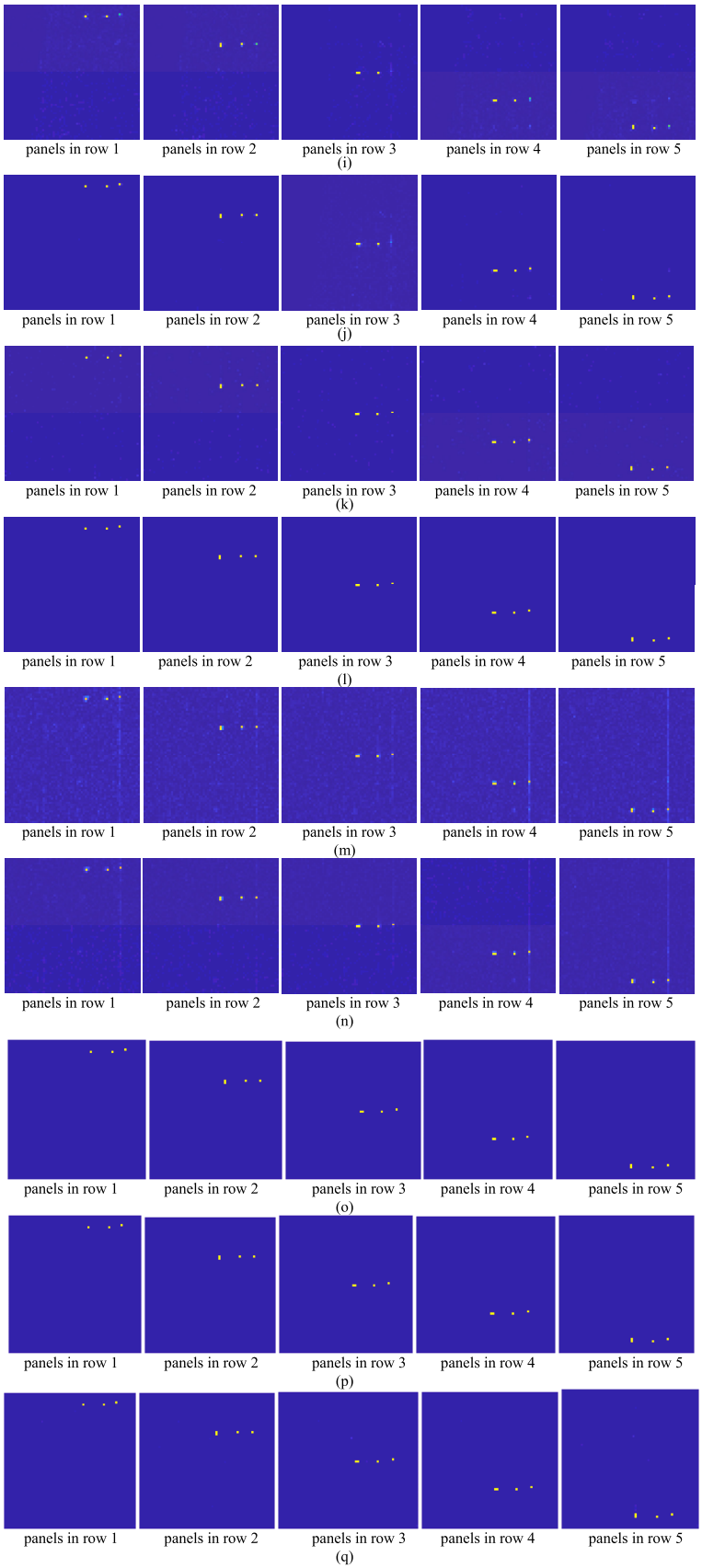


Fig. 6. *(Continued.)* Target abundance detection maps of various target detectors for HYDICE data scene. Detection maps of (i) KTCIMF, (j) IKTCIMF, (k) HCEM, (l) IHCEM, (m) ECEM, (n) IECEM, (o) DS-BA-KTCIMF, (p) LRaSMD-BA-KTCIMF, and (q) CDA-BA-KTCIMF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROWS DETECTION RESULTS $\mathbf{d} = \mathbf{p}_1$

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time (s)
$\mathbf{p}_1$	CEM	0.9989	0.39297	0.0084	1.39187	0.9905	0.38457	46.77657	1.38347	0.98
	ICEM	0.99951	0.62261	0.00903	1.62212	0.99048	0.61358	68.96467	1.6131	6.37
	KCEM	**1**	0.85821	0.00572	1.85821	0.99428	0.85249	150.063	1.85249	21.43
	IKCEM	**1**	0.99455	0.00142	1.99455	0.99858	0.99313	701.3752	1.99313	43.55
	LCMV	0.9989	0.39373	0.0084	1.39263	0.9905	0.38533	46.86134	1.38423	4.53
	ILCMV	0.99902	0.41171	0.0083	1.41073	0.99072	0.40341	49.60899	1.40243	16.64
	TCIMF	0.9989	0.39373	0.0084	1.39263	0.9905	0.38533	46.86134	1.38423	0.94
	ITCIMF	0.99951	0.62011	0.00902	1.61962	0.99049	0.61109	68.74083	1.6106	6.11
	KTCIMF	**1**	0.85799	0.00573	1.85799	0.99427	0.85225	149.7095	1.85225	21.47
	IKTCIMF	**1**	0.99455	0.00142	1.99455	0.99858	0.99313	699.8937	1.99313	42.89
	HCEM	**1**	0.94635	0.00103	1.94635	0.99897	0.94532	915.2311	1.94532	1.09
	IHCEM	**1**	0.99867	**0.00017**	**1.99867**	**0.99983**	**0.9985**	**6016.066**	**1.9985**	1.77
	ECEM	**1**	0.87933	0.01814	1.87933	0.98187	0.8612	48.48817	1.8612	1.40
	IECEM	**1**	0.8835	0.00856	1.8835	0.99144	0.87494	103.1769	1.87494	2.87
	DS-BA-KTCIMF	**1**	**0.9967**	0.0050	1.9967	0.9950	0.9917	199.3333	1.9917	283.5625
	LRaSMD-BA-KTCIMF	**1**	**0.9967**	0.0050	1.9967	0.9950	0.9917	199.3333	1.9917	125.1055
	CDA-BA-KTCIMF	**1**	**0.9967**	0.0050	1.9967	0.9950	0.9917	198.8475	1.9917	105.7600

TABLE II

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROW DETECTION RESULTS $\mathbf{d} = \mathbf{p}_2$

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time (s)
$\mathbf{p}_2$	CEM	0.99976	0.77665	0.01691	1.7764	0.98285	0.75974	45.93642	1.7595	0.87
	ICEM	0.99976	0.77202	0.01056	1.77177	0.9892	0.76146	73.13556	1.76122	1.78
	KCEM	**1**	0.89941	0.00493	1.89941	0.99508	0.89449	182.6217	1.89449	21.72
	IKCEM	**1**	0.99183	0.00161	1.99183	0.99839	0.99022	617.1923	1.99022	44.14
	LCMV	0.99976	0.77643	0.01788	1.77618	0.98188	0.75855	43.43648	1.75831	4.53
	ILCMV	0.99976	0.76967	0.00918	1.76943	0.99057	0.76049	83.80597	1.76025	16.64
	TCIMF	0.99976	0.77643	0.01788	1.77618	0.98188	0.75855	43.43648	1.75831	0.86
	ITCIMF	0.99976	0.7714	0.01061	1.77116	0.98914	0.76079	72.67797	1.76055	1.68
	KTCIMF	**1**	0.8991	0.00494	1.8991	0.99506	0.89416	182.0413	1.89416	21.96
	IKTCIMF	**1**	0.99175	0.00162	1.99175	0.99838	0.99013	613.3247	1.99013	44.02
	HCEM	**1**	0.96855	0.00108	1.96855	0.99893	0.96747	900.974	1.96747	0.86
	IHCEM	**1**	**0.99898**	**0.00016**	**1.99898**	**0.99985**	**0.99883**	**6445.032**	**1.99883**	1.67
	ECEM	**1**	0.93248	0.01473	1.93248	0.98527	0.91775	63.31758	1.91775	1.27
	IECEM	**1**	0.96523	0.00712	1.96523	0.99288	0.95811	135.5852	1.95811	2.70
	DS-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	282.2028
	LRaSMD-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	127.1160
	CDA-BA-KTCIMF	1	0.9963	0.0050	1.9963	0.9950	0.9912	197.9920	1.9912	125.3686
	CDA-BA-KTCIMF	1	0.9938	0.0050	1.9938	0.9950	0.9887	196.9213	1.9887	107.5967

TABLE III

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROW DETECTION RESULTS $\mathbf{d} = \mathbf{p}_3$

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time (s)
$\mathbf{p}_3$	CEM	0.99959	0.68775	0.0108	1.68734	0.98879	0.67695	63.66886	1.67654	0.82
	ICEM	0.99959	0.67238	0.00833	1.67198	0.99126	0.66405	80.67951	1.66364	1.74
	KCEM	**1**	0.94239	0.00453	1.94239	0.99547	0.93786	207.8502	1.93786	21.68
	IKCEM	**1**	0.99814	0.00101	1.99814	0.99899	0.99714	991.2046	1.99714	44.53
	LCMV	0.99919	0.55558	0.01257	1.55476	0.98661	0.54301	44.19513	1.54219	4.53
	ILCMV	0.99935	0.6037	0.00856	1.60305	0.99079	0.59514	70.52582	0.99935	16.64
	TCIMF	0.99919	0.55558	0.01257	1.55476	0.98661	0.54301	44.19513	1.54219	0.86
	ITCIMF	0.99902	0.57284	0.00893	1.57186	0.99009	0.56391	64.14782	1.56293	1.69
	KTCIMF	**1**	0.84611	0.00389	1.84611	0.99611	0.84222	217.6758	1.84222	22.65
	IKTCIMF	**1**	0.87828	0.00393	1.87828	0.99607	0.87435	223.5373	1.87435	44.09
	HCEM	**1**	0.97842	0.00092	1.97842	0.99909	0.9775	1069.31	1.9775	0.82
	IHCEM	**1**	**0.99986**	**0.00016**	**1.99986**	**0.99984**	**0.9997**	**6328.222**	**1.9997**	1.58
	ECEM	**1**	0.90331	0.0136	1.90331	0.9864	0.88971	66.41001	1.88971	1.27
	IECEM	**1**	0.9457	0.00678	1.9457	0.99322	0.93892	139.5245	1.93892	2.57
	DS-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	283.1608
	LRaSMD-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	126.7084
	CDA-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9912	197.9920	1.9912	115.3024

produced by IHCEM in Fig. 7(f), which used **d** calculated by the full set of 85 pixels, their detection maps looked nearly the same. However, the quantitative results of IHCEM in Table VI were significantly better than that of IRTS-HCEM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

TABLE IV

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROW DETECTION RESULTS $\mathbf{d} = \mathbf{p}_4$

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time (s)
$\mathbf{p}_4$	CEM	0.99976	0.75321	0.01038	1.75297	0.98938	0.74284	72.58495	1.74259	0.81
	ICEM	0.99967	0.70543	0.00794	1.7051	0.99173	0.69748	88.79985	1.69716	1.705
	KCEM	**1**	0.92367	0.00438	1.92367	0.99562	0.91929	210.7875	1.91929	22.05
	IKCEM	**1**	0.99208	0.00208	1.99208	0.99793	0.99	478.1089	1.99	44.14
	LCMV	0.99976	0.75168	0.01128	1.75143	0.98848	0.7404	66.63803	1.74015	4.53
	ILCMV	0.99967	0.73603	0.00713	1.7357	0.99255	0.7289	103.2731	0.99967	16.64
	TCIMF	0.99976	0.75168	0.01128	1.75143	0.98848	0.7404	66.63803	1.74015	0.85
	ITCIMF	0.99967	0.70167	0.00799	1.70134	0.99168	0.69368	87.77446	1.69335	1.67
	KTCIMF	**1**	0.92247	0.00444	1.92247	0.99556	0.91803	207.7628	1.91803	22.01
	IKTCIMF	**1**	0.99148	0.00214	1.99148	0.99786	0.98934	463.3103	1.98934	44.62
	HCEM	**1**	0.96669	0.00105	1.96669	0.99896	0.96564	925.0593	1.96564	0.82
	IHCEM	**1**	**0.99929**	**0.00016**	**1.99929**	**0.99984**	**0.99912**	**6245.531**	**1.99913**	1.59
	ECEM	**1**	0.93807	0.01617	1.93807	0.98383	0.9219	58.02029	1.9219	1.25
	IECEM	**1**	0.98067	0.00795	1.98067	0.99205	0.97272	123.3083	1.97272	2.68
	DS-BA-KTCIMF	1.0000	0.9950	0.0050	1.9950	0.9950	0.9900	199.0000	1.9900	283.4540
	LRaSMD-BA-KTCIMF	1.0000	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	127.6413
	CDA-BA-KTCIMF	1.0000	0.9963	0.0050	1.9963	0.9950	0.9912	199.1527	1.9912	108.1894

TABLE V

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR FIVE ROW DETECTION RESULTS USING $\mathbf{d} = \mathbf{p}_5$

d	detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time (s)
$\mathbf{p}_5$	CEM	0.99967	0.65011	0.01032	1.64979	0.98935	0.63979	62.98334	1.63947	0.87
	ICEM	0.99967	0.64252	0.00787	1.64219	0.9918	0.63465	81.65167	1.63432	1.71
	KCEM	**1**	0.937	0.00452	1.937	0.99548	0.93248	207.3922	1.93248	22.25
	IKCEM	**1**	0.99642	0.00135	1.99642	0.99865	0.99507	735.9106	1.99507	43.64
	LCMV	0.99967	0.64734	0.01112	1.64702	0.98856	0.63622	58.22983	1.6359	4.53
	ILCMV	0.99967	0.63809	0.00722	1.63777	0.99246	0.63088	88.42766	1.63055	16.64
	TCIMF	0.99967	0.64734	0.01112	1.64702	0.98856	0.63622	58.22983	1.6359	0.83
	ITCIMF	0.99967	0.64255	0.00793	1.64223	0.99175	0.63462	81.04844	1.6343	1.73
	KTCIMF	**1**	0.93571	0.00458	1.93571	0.99542	0.93112	204.1687	1.93112	21.96
	IKTCIMF	**1**	0.99657	0.00133	1.99657	0.99867	0.99524	747.6159	1.99524	43.79
	HCEM	**1**	0.93866	0.00088	1.93866	0.99912	0.93778	1066.66	1.93778	0.84
	IHCEM	**1**	**0.99929**	**0.00016**	**1.99929**	**0.99984**	**0.99913**	**6093.244**	**1.99913**	1.61
	ECEM	**1**	0.91609	0.01598	1.91609	0.98402	0.9001	57.32338	1.9001	1.30
	IECEM	**1**	0.96342	0.00884	1.96342	0.99116	0.95458	108.9592	1.95458	2.59
	DS-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	285.1055
	LRaSMD-BA-KTCIMF	**1**	0.9963	0.0050	1.9963	0.9950	0.9913	199.2500	1.9913	127.9219
	CDA-BA-KTCIMF	**1**	0.9938	0.0050	1.9938	0.9950	0.9887	196.9213	1.9887	107.5967

TABLE VI

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FOR SAN DIEGO DETECTION RESULTS

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	time
CEM	0.98999	0.48008	0.03009	1.47007	0.95991	0.44999	15.95633	1.43998	3.82
ICEM	0.99945	0.5454	0.00776	1.54485	0.99169	0.53764	70.27471	1.53709	24.66
KCEM	0.99494	0.60595	0.01041	1.60089	0.98453	0.59555	58.21443	1.59048	49.68
IKCEM	0.99997	**0.7763**	0.00366	**1.77627**	0.99631	**0.77264**	212.1621	**1.77261**	464.69
HCEM	0.99832	0.53044	0.00426	1.52876	0.99406	0.52618	124.5164	1.5245	3.82
IHCEM	**1**	0.74167	**0.00094**	1.74167	**0.99906**	0.74073	**789.8467**	1.74073	22.38
ECEM	0.99329	0.46768	0.02723	1.46097	0.96607	0.44045	17.17688	1.43374	5.52
IECEM	0.99005	0.61048	0.027	1.60054	0.96305	0.58348	22.60964	1.57353	21.18

TABLE VII

AUC VALUES CALCULATED FROM FIG. 8 FOR IRTS-CEM USING VARIOUS PERCENTAGES OF TRAINING SAMPLES

d	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	time
10%	0.99877	0.49557	0.02394	1.49434	0.97483	0.47163	20.7012	1.4704	38.8
20%	0.99462	0.49883	0.02075	1.49345	0.97387	0.47809	24.04579	1.4727	39.7
30%	0.99936	**0.53468**	0.01355	**1.53404**	0.98581	**0.52114**	39.46881	**1.5205**	40.76
40%	0.9994	0.52239	0.01199	1.52179	0.98742	0.5104	43.57929	1.5098	41.26
50%	**0.99942**	0.51661	**0.01143**	1.51604	**0.98799**	0.50518	**45.18226**	1.5046	40.87

using 50% training sampling rate in Table VIII. These experiments demonstrated that comparison using visual inspection is generally unreliable.

Fig. 11(a)–(e) shows the detection maps of IRTS-ECEM where, except the detection map in Fig. 11(a), all the detection maps in Fig. 11(b)–(e) looked nearly the same. However, when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

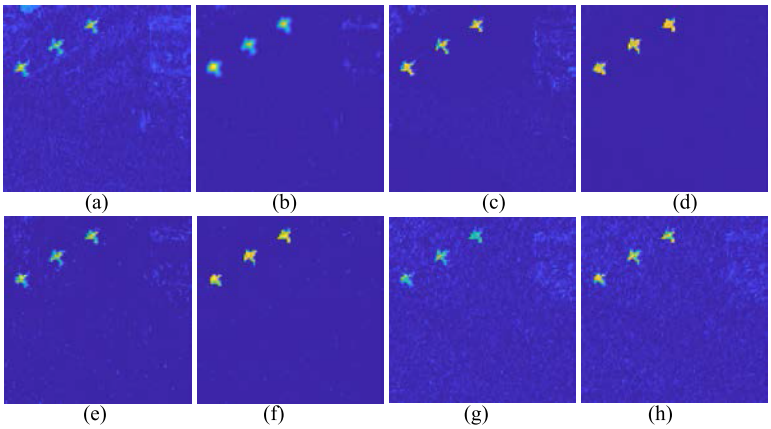


Fig. 7. Target detection maps of CEM, ICEM, KCEM, HCEM, IHCEM, ECEM, and IECEM for San Diego data scene. (a) CEM, (b) ICEM, (c) KCEM, (d) IKCEM, (e) HCEM, (f) IHCEM, (g) ECEM, and (h) IECEM.

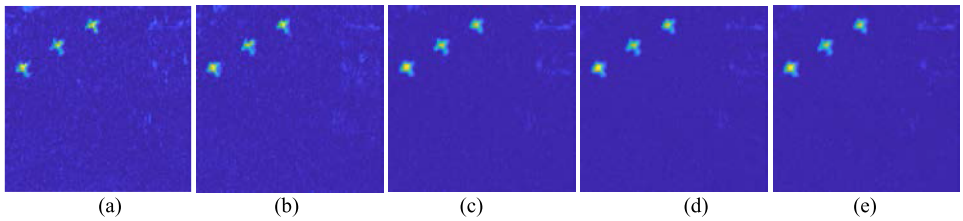


Fig. 8. Target detection maps of IRTS-CEM using various percentages of training samples for San Diego data scene. (a) 10%, (b) 20%, (c) 30%, (d) 40%, and (e) 50%.

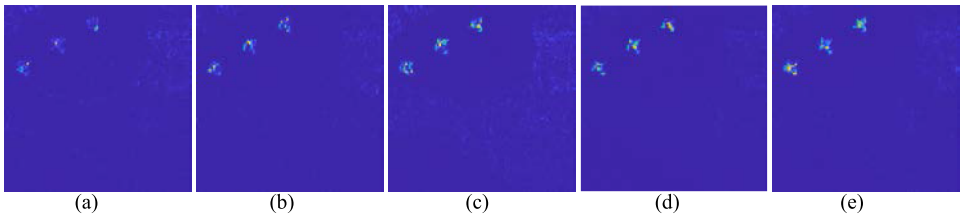


Fig. 9. Target detection maps of IRTS-KCEM using various percentages of training samples for San Diego data scene. (a) 10%, (b) 20%, (c) 30%, (d) 40%, and (e) 50%.

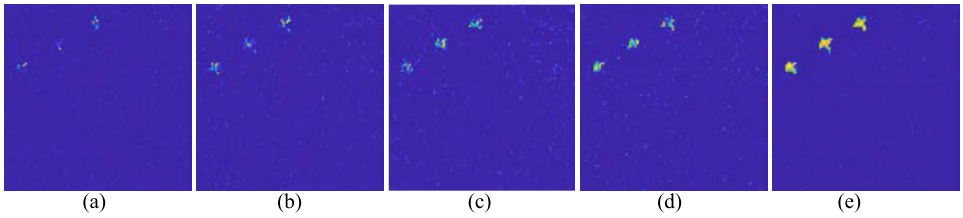


Fig. 10. Target detection maps of IRTS-HCEM using various percentages of training samples for San Diego data scene. (a) 10%, (b) 20%, (c) 30%, (d) 40%, and (e) 50%.

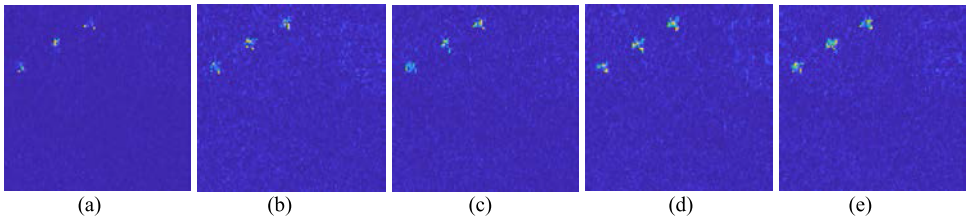


Fig. 11. Target detection maps of IRTS-ECEM using various percentages of training samples for San Diego data scene. (a) 10%, (b) 20%, (c) 30%, (d) 40%, and (e) 50%.

comparing the detection maps of IRTS-ECEM in Fig. 11(b)–(e) to the detection maps of IRTS-ECEM in Fig. 10(b)–(e), IRTS-HCEM performed better than IECEM. Interestingly, for the case of using 50% training sample rate, IRTS-HCEM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
AUC VALUES CALCULATED FROM FIG. 9 FOR IRTS-KCEM USING VARIOUS PERCENTAGES OF TRAINING SAMPLES

d	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	Time
10%	0.93796	0.11097	0.00654	1.04893	0.93142	0.10442	16.95675	1.04238	502.27
20%	0.98308	0.20828	0.00564	1.19136	0.97743	0.20264	36.90308	1.18571	514.04
30%	0.97309	0.31103	0.00895	1.28411	0.96414	0.30208	34.76716	1.27517	506.14
40%	0.97962	0.30293	**0.00408**	1.28255	0.97554	0.29885	**74.21068**	1.27847	508.85
50%	**0.99758**	**0.36912**	0.00561	**1.36671**	**0.99197**	**0.36351**	65.79715	**1.3611**	515.9

TABLE IX
AUC VALUES CALCULATED FROM FIG. 10 FOR IRTS-HCEM USING VARIOUS PERCENTAGES OF TRAINING SAMPLES

d	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	time
10%	0.81432	0.09326	**0.00149**	0.90758	0.81284	0.09178	62.80135	0.9061	35.99
20%	0.87691	0.1995	0.00255	1.07641	0.87437	0.19695	78.29474	1.07386	37.55
30%	0.93441	0.30317	0.00323	1.23758	0.93118	0.29994	93.94701	1.23435	36.92
40%	0.95373	0.33568	0.00304	1.28941	0.95069	0.33264	110.3475	1.28637	36.79
50%	**0.97189**	**0.39403**	0.00283	**1.36593**	**0.96907**	**0.39121**	**139.4811**	**1.3631**	37.44

TABLE X
AUC VALUES CALCULATED FROM FIG. 11 FOR IRTS-ECEM USING VARIOUS PERCENTAGES OF TRAINING SAMPLES

d	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{TDBS}	AUC_{SNPR}	AUC_{ODP}	time
10%	0.79879	0.15232	**0.01407**	0.95111	0.78472	0.13825	10.82566	0.93704	52.66
20%	0.90338	0.28396	0.02779	1.18734	0.87559	0.25618	10.21931	1.15955	53.65
30%	0.90274	0.28585	0.02323	1.18859	0.87952	0.26263	12.30746	1.16537	53.35
40%	**0.95999**	0.40197	0.0267	1.36196	**0.93329**	0.37527	15.05543	1.33526	53.72
50%	0.95510	**0.42322**	0.0273	**1.37832**	0.9278	**0.39593**	**15.50555**	**1.35103**	53.75

did much better than IRTS-KCEM in terms of detecting three airplanes. To further conduct the quantitative analysis Table X calculated the AUC values by eight detection measures from Fig. 11 where the best results were bold-faced where all the AUC values increased monotonically as the training sample rate was increased from 10% to 50% except for two cases, 10% with best value of $AUC_{(F,\tau)}$ and 40% with best values of $AUC_{(D,F)}$ and AUC_{BS}. If we further compare IRTS-ECEM using 50% training sampling rate to IRTS-HCEM using 50% training sampling rate, IRTS-HCEM did better than IECEM on five of eight AUC values. Specifically, IRTS-ECM did better than IRTS-HCEM in target detection with better $AUC_{(D,\tau)}$, AUC_{TD}, and AUC_{TDBS} values, while IRTS-HCEM did better than IRTS-ECEM in BS with better $AUC_{(D,F)}$, $AUC_{(F,\tau)}$, AUC_{BS}, AUC_{SNPR}, and AUC_{ODP} values.

C. Discussion

For a target detector to work effectively, two issues must be addressed: TD, which must be enhanced, and BS, which includes noise removal. Traditional target detectors are statistical detectors derived from the likelihood ratio test (LRT) based on the signal detection model to enhance TD via various particularly designed matched filters without taking care of BS issue [3], [38], [39]. Until recently, the BKG issue arising in HTD has begun to receive considerable interest due to very complicated BKG caused by many unknown and unidentified signal sources, which can be uncovered by a hyperspectral imaging system using hundreds of contiguous spectral bands. By realizing the fact that BKG significantly affects target detection, numerous recent works reported in the literature have been devoted to dealing with BKG. Specifically, DL-based methods are major trends to address the BKG issue. Just name a few examples. Li et al. [40] developed a paradigm that combines low-rank representation (LRR) model with DL techniques, called learning disentangled priors (LDP) to capture complete priors for effectively modeling the BKG. Gao et al. [41] developed a spectral aggregation and separation network for BKG separation. Li et al. [42] considered a joint overcomplete dictionary for sparse representation of targets and a global BKG dictionary for the collaborative representation of BKG. Tian et al. [43] derived an approach for target detection learning faithful BKG representations via orthogonal subspace-guided variational autoencoder. Shi et al. [44] presented a novel target-driven iterative AE (ITAE) framework by including an iterative system that repeatedly trains the same TAE and adds the weighted CEM-detection map back to the current data matrix for the next iteration. Unfortunately, hyperspectral image BKG is generally very sophisticated, and finding creditable and reliable training samples for DL methods is generally very difficult. This leads to the fact that the BKG estimated or constructed by such DL-based methods using training samples is usually not representative so that the BKG removal is not always effective. The CEM-generalized versions proposed in this article, IKCEM, IKTCIMF, IHCEM, and IECEM, do not require BKG training samples for BS. Instead, they only need to invert the correlation matrix $\mathbf{R}$ utilizing kernel functions and iterative processes to achieve BS. As experiments demonstrated in Sections VIII-B and VIII-C, these CEM-generalized methods achieved very impressive TD as well as cleaned up BKG. Since all the abovementioned methods [40], [41], [42], [43], [44], [45] also used the same San Diego airport data in Fig. 5 for experiments

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
ADVANTAGES AND DISADVANTAGES OF CEM, ICEM, KCEM, IKCEM, TCIMF, AND BA-TCIMF

detector	advantages	disadvantages
CEM	• requires only **d** • no need of background knowledge • does not need any model • simple and easy to implement	• sensitive to **d** • its effectiveness limited by linearity • is a spectral detector without taking care of target spatial information
ICEM	• requires only **d** • no need of background knowledge • includes a spatial filter to capture spatial information from the CEM-detection map and then feed it back and added to augment the currently being processed MR image cube to a new data cube in an iterative process • improves robustness of **d** to detection by updating and augmenting MR image cube via an iterative process	• Although ICEM reduces the sensitivity to its use of **d**, its detection performance is still limited to and determined by precise knowledge of **d.** • The effectiveness of ICEM is still limited by its linearity.
KCEM	• requires only **d** • no need of background knowledge • resolves nonlinear mixing arising in target pixels	• Due to its use of a nonlinear kernel, KCEM is very sensitive to **d**. As a result, KCEM produces many falsely alarmed pixels. • is still a spectral detector and does not take care of spatial information among detected pixels. • requires more computing time
IKCEM	• Since IKCEM combines the strengths of both ICEM and KCEM, it not only can capture the spatial information among detected target pixels, but also can resolve issues of nonlinear mixing in target pixels to further reduce false alarm probability.	• IKCEM is a single-target nonlinear detector. It cannot detect multiple targets at the same time. • requires more computing time
LCMV	• requires only knowledge of multiple targets • extends CEM to multiple-target detection • extends CEM to multiple-class classification • suppresses global BKG	• its effectiveness limited by linearity • is a spectral detector without taking care of target spatial information
TCIMF	• requires only knowledge of multiple targets • extends CEM to multiple-target detection • annihilates additional unwanted signal sources • suppresses global BKG	• its effectiveness limited by linearity • finds an appropriate set of undesired signal sources to be annihilated
BA-TCIMF	• extends CEM to multiple-target detection • annihilates additional unwanted signal sources • suppresses global BKG • requires LRaSMD model to perform target BKG suppression	• its effectiveness limited by linearity • finds an appropriate set of undesired signal sources to be annihilated • determines the order of LRaSMD model.

to justify their claims by 3-D ROC analysis, those who are interested in comparison with the work presented in this article are encouraged to refer these works.

Finally, since CEM derived many generalizations with various variants, Table XI summarizes the advantages and disadvantages of CEM, ICEM, KCEM, IKCEM, TCIMF, and BA-TCIMF where KLCMV, KTCIMF, and BA-KTCIMF are not included because they have the same advantages and disadvantages of using kernels as KCEM does.

D. Comparison of CEM With DL or Low-Rank and Sparse Representation-Based Methods

Recently, DL and low-rank and sparse representation (LRSR)-based methods have received considerable interest in HTD. There are reasons that this article does not compare CEM with DL- or LRSR-based methods.

1) CEM does not need any model, while DL- and LRSR-based methods do.
2) CEM only requires the knowledge of the desired target signatures, which can be obtained by prior knowledge such as spectral library or by visual inspection. Compared to CEM, DL- and LRSR-based methods require models to obtain target knowledge.
3) CEM does not require training samples to train its filter, whereas DL- and LRSR-based methods need a large number of network parameters to be tuned for training.
4) CEM is simple and very easy to implement as opposed to DL- and LRSR-based methods that have very high computational complexity and also excessive computing time.
5) CEM is a statistical detector that makes use of a matched filter via desired target signatures to enhance TD while inverting correlation matrix **R** so as to achieve global BS. By contrast, DL- and LRSR-based methods are training sampling-based methods. A key issue arising in these methods is how to select a creditable and reliable training sample pool to train their networks. This is specifically troublesome with obtaining training samples from complicated BKG as shown in [37, Fig. 18].

With all things considered above, comparing CEM to DL- and LRSR-based methods is not appropriate like comparing an apple to an orange.

IX. CONCLUSION

CEM is an effective detection technique for HTD. Due to its simple structure, many of them have taken it for granted, and very few paid attention to its design rationale and potential

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

in applications. The main ideas of this article are itemized as follows.

1) This article provides a mathematical theory of CEM.
2) Based on its mathematical foundation a new version of CEM, SBR-based CEM can be further derived to show that the MV of CEM is exactly the reciprocal of its SBR.
3) Specifically, this article also proves why CEM works mathematically.
4) This article also shows the versatility of CEM by deriving several new generalizations for HTD, iterative IKCEM, IHCEM, IECEM, IRTS KCEM (IRTS-KCEM), IRTS HCEM (IRTS-HCEM), and IRTS ECEM (IRTS-ECEM), all of which are new and have not explored for HTD. These algorithms turn out to perform better than many current HTD algorithms compared in [3].

As also shown in the experiments conducted in Section VIII, the best performance was achieved by IHCEM with $\mathbf{d}$ calculated by complete GT for both datasets. However, when the training samples are used to calculate $\mathbf{d}$ for the San Diego airport data, IRTS-KCEM turns out to be the winner and performs significantly better than IHCEM. This indicated that HCEM is very sensitive to its use of the knowledge of $\mathbf{d}$. This evidence was also justified by the experiments in Section VIII [45].

Finally, as for future research directions, CEM has also found its role in many areas, anomaly detection [37], [46], band selection [47], [48], [49], [50], [51], image classification [11], [14], [19], [22], [23], [52], multispectral imaging [53], medical imaging [54], [55], [56], [57], and many other applications yet to explore, such as geological mineral extraction and identification, pesticide pollution in agriculture [58], plant disease detection in forestry [59], food safety and inspection, and environmental monitoring.

ACKNOWLEDGMENT

The author would like to express his sincere gratitude to Dr. Kenneth Yeonkang Ma with the Remote Sensing Signal and Image Processing Laboratory at the University of Maryland, Baltimore County, for his help in implementing experiments presented in this article.

REFERENCES

[1] C.-I Chang, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification*. New York, NY, USA: Kluwer Academic, 2003.
[2] C.-I Chang, *Real-Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection*. New York, NY, USA: Springer, 2016.
[3] C.-I Chang, "Hyperspectral target detection: Hypothesis testing, signal-to-noise ratio, and spectral angle theories," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5505223.
[4] C.-I Chang, "Hyperspectral anomaly detection: A dual theory of hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511720.
[5] J. C. Harsanyi and C.-I Chang, "Hyperspectral image classification and dimensionality reduction: An orthogonal subspace projection approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 4, pp. 779–785, Jul. 1994.
[6] J. C. Harsanyi, "Detection and classification of subpixel spectral signatures in hyperspectral image sequences," Ph.D. dissertation, Dept. Elect. Eng., Univ. Maryland, Baltimore County, MD, USA, Aug. 1993.
[7] W. Farrand, "Mapping the distribution of mine tailings in the Coeur D'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.
[8] C.-I Chang, "Target signature-constrained mixed pixel classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 5, pp. 1065–1081, May 2002.
[9] B. D. Van Veen and K. M. Buckley, "Beamforming: A versatile approach to spatial filtering," *IEEE ASSP Mag.*, vol. 5, no. 2, pp. 4–24, Apr. 1988.
[10] H. Tang. (2014). *DOA Estimation Based on MUSIC Algorithm*. [Online]. Available: http://www.diva-portal.org/smash/get/diva2:724272/FULLTEXT01.pdf
[11] B. Xue et al., "A subpixel target approach to hyperpsectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5093–5114, Sep. 2017.
[12] X. Yang, J. Chen, and Z. He, "Sparse–spatial CEM for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2184–2195, Jul. 2019.
[13] C.-I Chang, K. Y. Ma, C.-C. Liang, Y.-M. Kuo, S. Chen, and S. Zhong, "Iterative random training sampling spectral spatial classification for hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3986–4007, 2020.
[14] X. Shang, M. Song, and C.-I Chang, "An iterative random training sample selection approach to constrained energy minimization for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1625–1629, Sep. 2021.
[15] C.-I Chang, C.-C. Liang, and P. Hu, "Iterative convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Snes.*, vol. 61, 2023, Art. no. 5513526.
[16] X. Jiao and C.-I Chang, "Kernel-based constrained energy minimization," *Proc. SPIE*, vol. 6966, pp. 523–533, Apr. 2008.
[17] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.
[18] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, Jun. 2019, doi: 10.3390/rs111113.
[19] K. Y. Ma and C.-I Chang, "Kernel-based constrained energy minimization for hyperspectral mixed pixel classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510723.
[20] O. L. Frost III, "An algorithm for linearly constrained adaptive array processing," *Proc. IEEE*, vol. 60, no. 8, pp. 926–935, Aug. 1972.
[21] H. Ren and C.-I Chang, "Target-constrained interference-minimized approach to subpixel target detection for hyperspectral imagery," *Opt. Eng.*, vol. 39, no. 12, pp. 3138–3145, Dec. 2000.
[22] C. Yu, Y. Wang, M. Song, and C.-I Chang, "Class signature-constrained background-suppressed approach to band selection for classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 14–31, Jan. 2019.
[23] C. Yu, B. Xue, M. Song, Y. Wang, S. Li, and C.-I Chang, "Iterative target-constrained interference-minimized classifier for hyperspectral classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1095–1117, Apr. 2018.
[24] T. Wang, B. Du, and L. Zhang, "A kernel-based target-constrained interference-minimized filter for hyperspectral sub-pixel target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 626–637, Apr. 2013.
[25] J. Chen and C.-I Chang, "Background-annihilated target-constrained interference-minimized filter (TCIMF) for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540224.
[26] B. Picinbono, "On deflection as a performance criterion in detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 31, no. 3, pp. 1072–1081, Jul. 1995.
[27] H. V. Poor, *An Introduction to Signal Detection and Estimation*, 2nd ed., New York, NY, USA: Springer, 1991.
[28] T.-M. Tu, C.-H. Chen, and C.-I Chang, "A posteriori least squares orthogonal subspace projection approach to desired signature extraction and detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 1, pp. 127–139, Jan. 1997.
[29] C.-I Chang, "Chapter 3: Information-processed matched filters for hyperspectral target detection and classification," in *Hyperspectral Data Exploitation: Theory and Applications*, C.-I Chang, Ed., Hoboken, NJ, USA: Wiley, 2007, pp. 47–74.
[30] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990.
[31] S. Theodoridis and K. Koutroumbas, *Pattern Recognition*. Cambridge, MA, USA: Academic Press, 1999, p. 366.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.

[32] C.-I Chang, *Yperspectral Data Processing: Algorithm Design and Analysis*. Hoboken, NJ, USA: Wiley, 2013.
[33] T. Zhou and D. Tao, "GoDec: Randomized low-rank & sparsity matrix decomposition in noisy case," in *Proc. 28th Int. Conf. Mach. Learn.*, Bellevue, WA, USA, 2011, pp. 1–16.
[34] S. Chen, C.-I Chang, and X. Li, "Component decomposition analysis for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516222.
[35] C.-I Chang, "An effective evaluation tool for hyperspectral target detection: 3D receiver operating characteristic curve analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5131–5153, Jun. 2021.
[36] C.-I Chang, S. Chen, S. Zhong, and Y. Shi, "Exploration of data scene characterization and 3D ROC evaluation for hyperspectral anomaly detection," *Remote Sens.*, vol. 16, no. 1, p. 135, Dec. 2023, doi: 10.3390/rs16010135.
[37] C.-I Chang, "Target-to-anomaly conversion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540428.
[38] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.
[39] D. Manolakis, E. Truslow, M. Pieper, T. Cooley, and M. Brueggeman, "Detection algorithms in hyperspectral imaging systems," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 24–33, Jan. 2014.
[40] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, 2024.
[41] H. Gao, Y. Zhang, Z. Chen, F. Xu, D. Hong, and B. Zhang, "Hyperspectral target detection via spectral aggregation and separation network with target band random mask," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515516, doi: 10.1109/TGRS.2023.
[42] C. Li, D. Zhu, C. Wu, B. Du, and L. Zhang, "Global overcomplete dictionary-based sparse and nonnegative collaborative representation for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5513214, doi: 10.1109/TGRS.2024.3381719.
[43] Q. Tian, C. He, Y. Xu, Z. Wu, and Z. Wei, "Hyperspectral target detection: Learning faithful background representations via orthogonal subspace-guided variational autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516714, doi: 10.1109/TGRS.2024.3393931.
[44] Y. Shi, X. Li, and S. Chen, "Target-driven iterative autoencoder for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2023, Art. no. 5505518, doi: 10.1109/TGRS.2023.3346923.
[45] C.-I. Chang, "Effective anomaly space for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526624.
[46] C.-I Chang, "Constrained energy minimization anomaly detection for hyperspectral imagery via dummy variable trick," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5517119.
[47] X. Geng, K. Sun, and L. Ji, "Band selection for target detection in hyperspectral imagery using sparse CEM," *Remote Sens. Lett.*, vol. 5, no. 12, pp. 1022–1031, Dec. 2014.
[48] C.-I Chang and S. Wang, "Constrained band selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 6, pp. 1575–1585, Jun. 2006.
[49] L. Wang et al., "Band subset selection for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4887–4898, Sep. 2017.
[50] L. Wang, H.-C. Li, B. Xue, and C.-I Chang, "Constrained band subset selection for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2032–2036, Nov. 2017.
[51] X. Shang et al., "Target-constrained interference-minimized band selection for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6044–6064, Jul. 2021.
[52] M. Song, X. Shang, and C.-I Chang, "3-D receiver operating characteristic analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8093–8115, Nov. 2020.
[53] C.-I Chang et al., "A generalized constrained energy minimization approach to subpixel target detection for multispectral imagery," *Opt. Eng.*, vol. 39, no. 5, pp. 1275–1281, May 2000.
[54] H.-M. Chen et al., "A hyperspectral imaging approach to white matter hyperintensities detection in brain magnetic resonance images," *Remote Sens.*, vol. 9, no. 11, p. 1174, 2017, doi: 10.3390/rs9111174.
[55] C. C. Chen et al., "An iterative mixed pixel classification for brain tissues and white matter hyperintensity in magnetic resonance imaging," *IEEE Access*, vol. 7, pp. 124674–124687, 2019.
[56] C.-I Chang, C.-C. C. Chen, J. W. Chai, and H.-M. Chen, "Iterative analyzing method for a medical image," U.S. Patent 1 06 217 22B2, Apr. 14, 2020. [Online]. Available: https://patents.google.com/patent/US10621722B2/en
[57] Y.-C. Chang et al., "Detection of periventricular white matter hyperintensities in magnetic resonance imaging by hyperspectral subpixel detection," *IEEE Access*, 2024.
[58] K.-Y. Ma, Y.-M. Kuo, Y.-C. Ouyang, and C.-I Chang, "Improving pesticide residues detection using band prioritization and constrained energy minimization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Fort Worth, TX, USA, Jul. 2017, pp. 4802–4805.
[59] Y.-M. Kuo et al., "Quality inspection of agriculture products using target detection on hyperspectral synthetic image," in *Proc. IEEE/GRSS Int. Geosci. Remote Sens. Symp. (IGARSS)*, Valencia, Spain, Jul. 2018, pp. 23–27.



Chein-I Chang (Life Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been a Chang Jiang Scholar Chair Professor and the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016, and is also a Yushan Scholar Chair Professor at National Cheng Kung University, Tainan, Taiwan. In addition, he has been a Chair Professor at National Yang Ming Chiao Tung University, Hsinchu, Taiwan, since 2019. He is currently a Professor with the University of Maryland at Baltimore County (UMBC), Baltimore, MD, USA. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing: Endmember Finding and Anomaly Detection* (Springer, 2016), *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017), and a forthcoming book, *Statistical Hyperspectral Target and Anomaly Detection: Signal and Image Processing Perspectives* (Wiley, 2025). He has edited three books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006), *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007), and *Advances in Hyperspectral Image Processing Techniques* (Wiley, 2023). In addition, he also co-edited with A. Plaza a book on *High Performance Computing in Remote Sensing* (CRC Press, 2007). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE and is serving as an Associate Editor for *Remote Sensing* and IEEE TRANSACTION ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:09:33 UTC from IEEE Xplore. Restrictions apply.